

Automatic Speech Recognition using Advanced Deep Learning Approaches: A survey

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ARTICLE INFO

Keywords:

Automatic speech recognition

Deep transfer learning

Transformers

Federated learning

Reinforcement learning

ABSTRACT

Recent advancements in deep learning (DL) have posed a significant challenge for automatic speech recognition (ASR). ASR relies on extensive training datasets, including confidential ones, and demands substantial computational and storage resources. Enabling adaptive systems improves ASR performance in dynamic environments. DL techniques assume training and testing data originate from the same domain, which is not always true. Advanced DL techniques like deep transfer learning (DTL), federated learning (FL), and deep reinforcement learning (DRL) address these issues. DTL allows high-performance models using small yet related datasets, FL enables training on confidential data without dataset possession, and DRL optimizes decision-making in dynamic environments, reducing computation costs.

This survey offers a comprehensive review of DTL, FL, and DRL-based ASR frameworks, aiming to provide insights into the latest developments and aid researchers and professionals in understanding the current challenges. Additionally, Transformers, which are advanced DL techniques heavily used in proposed ASR frameworks, are considered in this survey for their ability to capture extensive dependencies in the input ASR sequence. The paper starts by presenting the background of DTL, FL, DRL, and Transformers and then adopts a well-designed taxonomy to outline the state-of-the-art (SOTA) approaches. Subsequently, a critical analysis is conducted to identify the strengths and weaknesses of each framework. Additionally, a comparative study is presented to highlight the existing challenges, paving the way for future research opportunities.

1. Introduction

1.1. Preliminary

Advancements in artificial intelligence (AI) have significantly improved human-machine interaction (HMI), especially with technologies that convert speech into executable actions. automatic speech recognition (ASR) emerges as a leading communication technology in HMI, extensively utilized by corporations and service providers for facilitating interactions through AI platforms like chatbots and digital assistants. Spoken language forms the core of these interactions, emphasizing the necessity for sophisticated speech processing in AI systems tailored for ASR. ASR technology encompasses the analysis of (i) acoustic, lexical, and syntactic aspects; and (ii) semantic understanding. The acoustic model (AM) processing includes speech coding [1], enhancement [2], source separation [2, 3], alongside securing speech via steganography [4–6] and watermarking [7–9]. These components are integral to audio analysis. On the other hand, the semantic model (SM), often identified as language model (LM) processing in literature, involves all natural language processing (NLP) techniques. This AI branch aims at teaching computers to understand and interpret human language, serving as the basis for applications like music information retrieval [10], sound file organization [11], audio tagging (AT), and event detection (ED) [12], as well as converting speech to text and vice versa [13], detecting hate speech [14], and cyberbullying [15]. Employing NLP across various domains enables AI models

to effectively comprehend and respond to human inputs, unveiling extensive research prospects in diverse sectors.

Recent advancements in ASR have been significantly propelled by the evolution of deep learning (DL) methodologies. An extensive range of DL models has been developed, demonstrating remarkable improvements and surpassing former SOTA achievements [16, 17]. Transformers, a notable innovation within these DL approaches, have become a cornerstone in advancing various NLP tasks, including ASR. Initially conceptualized for sequence-to-sequence (S2S) applications in NLP, their success is largely attributed to their adeptness at discerning long-range dependencies and complex patterns within sequential data. A hallmark of Transformer models is their utilization of an attention mechanism, which precisely focuses on specific portions of the input sequence during prediction tasks. This mechanism is particularly effective in ASR, facilitating the detailed modeling of contextual nuances and the interconnections among acoustic signals, essential for accurate transcription. Models such as the Transformer Transducer, Conformer, and end-to-end speech processing network (ESPNet), leveraging self-attention and parallel processing, have achieved leading performance in ASR tasks. Their robustness across diverse languages further underscores their capability to adapt to a wide range of linguistic features and acoustic variations, making Transformers an exceptionally promising option for enhancing ASR systems, surpassing the constraints of conventional models.

The integration of DL with its variants in ASR introduces substantial challenges, especially concerning its application in natural HMI. Despite DL's numerous advantages, it encounters various obstacles. The inherent complexity of DL models, which stems from their need for extensive training data to attain high performance, demands significant

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Abbreviations

AAC	automated audio captioning	FCF	feature correlation-based fusion	NT	negative transfer
AC	actor-critic	FedAvg	federated averaging	PESQ	perceptual evaluation of speech quality
AI	artificial intelligence	FedNST	federated noisy student training	QCNN	quantum CNN
AM	acoustic model	FL	federated learning	RER	relative error rate
APT	audio pyramid transformer	FMTL	federated multi-task learning	RNN	recurrent neural network
ASR	automatic speech recognition	FR	form recognition	RTF	real-time factor
AST	audio spectrogram transformer	GAN	generative adversarial network	S2S	sequence-to-sequence
AT	audio tagging	GLDPT	global-local dual-path Transformer	SARSA	state-action-reward-state-action
BERT	bidirectional encoder representations from transformers	HFL	horizontal federated learning	SCST	self-critical sequence training
CAFT	client adaptive federated training	HMI	human-machine interaction	SD	source domain
CER	character error rate	HMM	hidden Markov models	SE	speech enhancement
CLDNN	convolutional long-short term deep neural network	HSD	hate speech detection	SER	speech emotion recognition
CNN	convolutional neural network	ISTFT	inverse short-time Fourier transform	SM	semantic model
CS	code-switching	LLM	large language model	SNR	signal-to-noise ratio
CTC	connectionist temporal classification	LM	language model	SOTA	state-of-the-art
CV	computer vision	LPC	linear predictive coding	SS	speech security
DA	domain adaptation	LRF	low-rank factorization	SSAST	self-supervised audio spectrogram transformer
DDPG	deep deterministic policy gradient	LSTM	long short term memory	STFT	short-time Fourier transform
DDQN	double deep Q-network	MAE-AST	masked autoencoding audio spectrogram Transformer	SVD	singular value decomposition
DL	deep learning	mAP	mean average precision	SWBD	switchboard
DNN	deep neural network	MDP	Markov decision process	TBE	two-branch encoder
DRL	deep reinforcement learning	MFCC	Mel-frequency cepstral coefficient	TD	target domain
DSC	depthwise separable convolutions	MHSA	multi-head self-attention	TNet	Transformer network
DSLMM	domain-specific language modeling	ML	machine learning	TNR	true negative rate
DTL	deep transfer learning	MMD	maximum mean discrepancy	TPR	true positive rate
ED	event detection	MOS-LQO	mean opinion score-listening quality objective	TRUNet	transformer-recurrent-U network
ESPNNet	end-to-end speech processing network	MTL	multitask learning	TSTNN	Transformer-based neural network
		NLP	natural language processing	VFL	vertical federated learning
		non-IID	non-identically distributed	WER	word error rate
				WSJ	wall street journal

computational and storage resources [18]. Moreover, the issue of data scarcity in ASR reflects the inadequate quantities of training data available for exploiting complex DL algorithms effectively [19]. The paucity of annotated data further complicates the development of supervised DL-based ASR models. Additionally, the presumption that training and testing datasets originate from the same domain, possessing identical feature spaces and distribution characteristics, is often misguided. This mismatch challenges the practical deployment of DL models in real-world settings [20]. Thus, the performance of DL models may be compromised when faced with limited training datasets or discrepancies in data distribution between training and testing environments [21]. These challenges highlight the critical need for adaptive methodologies and improved data management approaches to fully harness the capabilities of DL in ASR systems.

In an effort to address existing challenges and increase the robustness and flexibility of ASR systems, novel DL methodologies have been introduced. These include deep

transfer learning (DTL), deep reinforcement learning (DRL), and federated learning (FL), which collectively aim at overcoming difficulties related to the transfer of knowledge, enhancing the generalization capabilities of models, and optimizing training processes. These innovative approaches significantly broaden the operational scope of conventional DL frameworks within the ASR field.

Figure 1 highlights critical areas in speech processing where DTL, DRL, and FL can be applied. Consequently, domains such as ASR, speech enhancement (SE), hate speech detection (HSD), and speech security (SS) are closely interconnected. ASR provides acoustic parameters to NLP for HSD task, which in turn provides semantic details to ASR. Additionally, ASR can be employed in the SS domain as a steganalytic process to verify the integrity of speech [4, 22]. Furthermore, ASR and SE can mutually offer performance feedback.

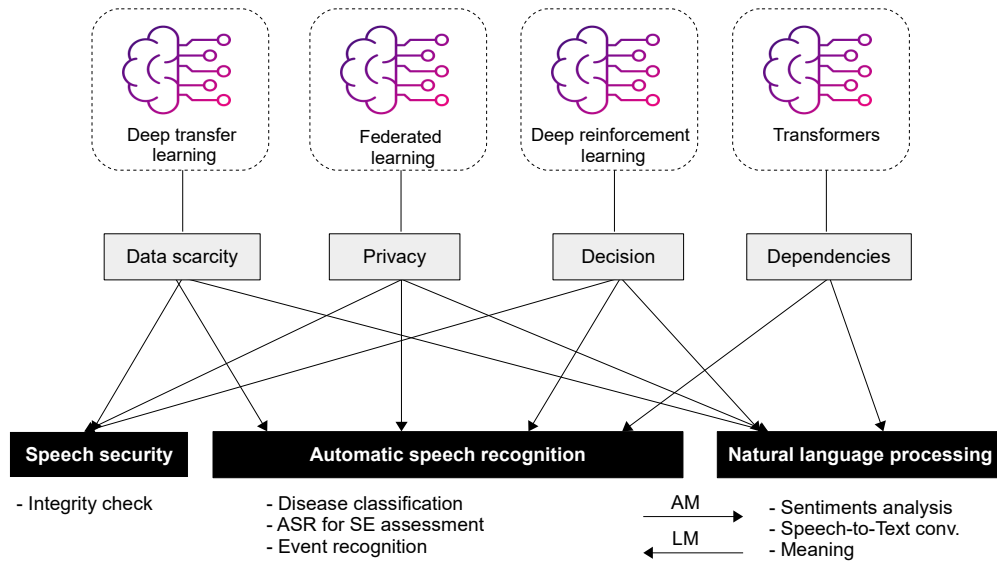


Figure 1: Summary of critical areas in speech processing where DTL, DRL, FL, and Transformers can be applied.

1.2. Contribution of the paper

This article offers an extensive examination of contemporary frameworks within advanced deep learning approaches, spanning the period from 2017 to 2023. These approaches include DTL, DRL, FL, and Transformers, all within the context of ASR. To the best of the authors' knowledge, there has been no prior research paper that has intricately explored and critically evaluated contributions in the aforementioned advanced DL-based ASR until now.

In recent years, numerous survey papers have been published to assess various aspects of ASR models. Some of these surveys concentrate on specific languages, such as Portuguese [23], Indian [24], Turkish [25], Arabic [26] and tonal languages (including Asian, Indo-European and African) [27]. Additionally, Abushariah et al.'s review emphasizes bilingual ASR [28]. On the non-specific language review front, specific areas within ASR have been targeted, including ASR using limited vocabulary [29], ASR for children [30], error detection and correction [31], and unsupervised ASR [32]. Systematic reviews with a focus on neural networks [33] and deep neural networks (DNNs) [34] have also been proposed. In another comprehensive review, Malik et al. [35] discussed diverse feature extraction methods, SOTA classification models, and some deep learning approaches. Recently, the authors presented an ASR review focused on DTL for ASR [36]. Table 1 presents a summary of the main contributions of the proposed DTL review compared to other existing DTL reviews/surveys.

This survey article offers several significant enhancements and additions compared to previous DTL surveys. Firstly, it consolidates works that utilize both DTL and advanced DL approaches, providing a comprehensive overview of their intersection. Secondly, it provides performance evaluation results of all considered approaches. Thirdly, it includes metrics and dataset reviews used in DTL models. Furthermore, it tackles ongoing challenges and consequently proposes future directions. The main contributions of this article can be summarized as follows:

- Presenting the background of advanced DL techniques including DTL, DRL, FL and Transformers. Describing the evaluation metrics and datasets employed for validating ASR approaches.
- Introducing a well-defined taxonomy categorizing ASR methodologies based on the domains of AM and LM.
- Identifying challenges and gaps in advanced DL-based ASR.
- Proposing future directions to enhance the performance of advanced DL-based ASR solutions and predicting the potential advancements in the field.

1.3. Review methodology

The methodology for the review is delineated in this segment, encompassing the search strategy and study selection. Inclusion criteria, comprising keyword alignment, creativity and impact, and uniqueness, are explicated, collectively influencing the formulation of the paper's quality assessment protocol. To locate and compile extant advanced DL-based ASR studies, a thorough search was executed on renowned publication databases recognized for hosting top-tier scientific research articles. The exploration encompassed Scopus and Web of Science. Keywords were extracted and organized from the initial set of references through manual analysis. Employing "theme clustering," these publications were sorted based on keywords found in the "Abstract," "Title," and "Authors keywords." The outcome of this process yielded the formulation of the following query:

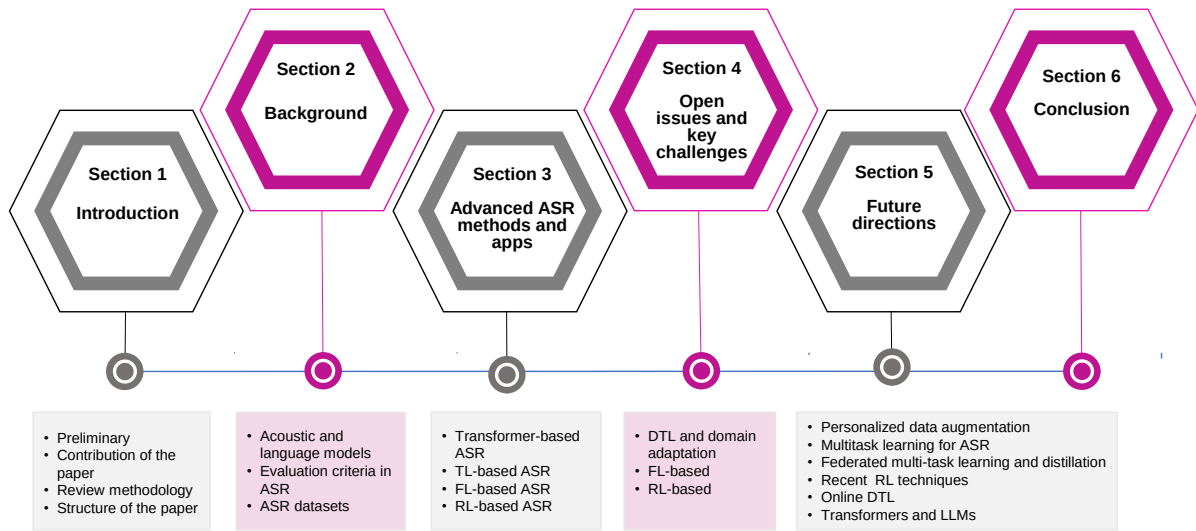
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References=FROM "Abstract" || "Title" || "Authors keywords" SELECT( Papers WHERE keywords= (ASR || NLP) & ( DTL || DRL || FL || Transformers)).
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The symbols || and & denote OR and AND logical operations, respectively. The evaluation of publications considered the innovation level in ASR, the study's quality, and the contributions and findings presented. This review

Table 1

Contribution comparison of the proposed contribution against other hand DTL review. The tick mark (✓) indicates that the specific field has been addressed, whereas the cross mark (X) means addressing the specific fields has been missed.

Refs	Year	Description of the survey/review	Advanced DL methods				Performances evaluation	Metrics	Dataset review	Current challe/Gaps	Future directions
			DRL	FL	DTL	Transf.					
[31]	2018	ASR review for errors detection and correction	X	X	X	X	X	✓	X	X	X
[34]	2019	Systematic review on DL-based speech recognition	X	X	X	X	X	X	X	X	X
[24]	2020	ASR survey for Indian languages	X	X	X	X	X	X	✓	X	✓
[23]	2020	ASR survey for Portuguese language	X	X	X	X	X	X	✓	✓	✓
[25]	2020	ASR survey for Turkish language	X	X	X	X	✓	X	✓	X	X
[35]	2021	ASR survey	X	X	X	X	X	X	X	✓	✓
[27]	2021	ASR survey for tonal languages	X	X	X	X	✓	X	✓	✓	✓
[32]	2022	Unsupervised ASR review	X	X	X	X	✓	X	X	✓	X
[30]	2022	ASR Systematic review for children	X	X	X	X	X	X	✓	X	X
[29]	2022	ASR survey for limited vocabulary	✓	X	X	X	X	X	X	X	✓
[26]	2022	ASR Systematic review for Arabic language	X	X	X	X	✓	X	X	✓	✓
[28]	2022	Bilingual ASR review	X	X	X	X	✓	X	✓	✓	X
[33]	2023	ASR survey on neural network techniques	X	X	✓	X	✓	✓	✓	✓	X
[36]	2023	ASR based on DTL review	X	X	✓	X	✓	✓	✓	✓	✓
Our	2024	ASR review on advanced DL techniques	✓	✓	✓	✓	✓	✓	✓	✓	✓

**Figure 2:** Survey roadmap: A guide for navigating paper sections and subsections.

exclusively encompassed research contributions that were published within the timeframe of 2016 to 2023.

1.4. Structure of the paper

This paper is structured into six sections. The current section provides an introduction to the paper. Section 2 providing background on AM and LM, and reviewing evaluation metrics and datasets utilized in ASR. Moving forward, Section 3 delves into a comprehensive review of recent advancements in ASR utilizing advanced DL approaches, including Transformers, DTL, FL and DRL. Sections 4 and 5 respectively address the existing challenges and future directions concerning advanced DL-based ASR. Finally, Section 6 presents concluding remarks summarizing the key findings of the paper. Figure 2 presents a structured roadmap, offering a comprehensive guide to assist readers in navigating through the various sections and subsections of the paper.

2. Background

2.1. Acoustic and language models

The AM is in charge of capturing the sound characteristics of different phonetic units. This involves generating statistical measures for characteristic vector sequences from the audio waveform. Various techniques, such as linear predictive coding (LPC), Cepstral analysis, filter-bank analysis, Mel-frequency cepstral coefficients (MFCCs), wavelet analysis, and others, can be used to extract these features [37]. In the processing stage, a decoder (search algorithm) uses the acoustic lexicon and LM to create the hypothesized word or phoneme. You can see the overall process illustrated in Figure 3.

LMs provide probabilities of sequences of words, crucial for ASR systems to predict the likelihood of subsequent words in a sentence [38]. A domain-specific LM is trained on text data from the target domain to capture its unique vocabulary and grammatical structures [39]. For n-gram models, this involves calculating the conditional probability

Table 2

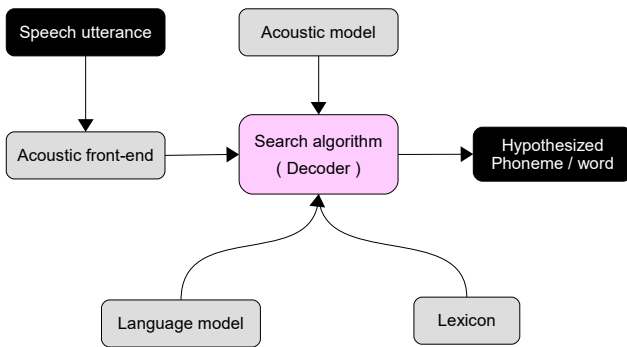
An overview of the metrics employed for evaluating ASR methods.

Metric	Formula	Description
WER	$\frac{S + D + I}{N} = \frac{S + D + I}{H + S + D}$	The word error rate (WER) serves as a frequently utilized metric to assess the performance of Automatic Speech Recognition (ASR). It is computed by determining the ratio of incorrectly recognized words to the overall number of processed words [17, 45, 46]. In the given context, I, D, S, H, and N denote the quantities of insertions, deletions, substitutions, hits, and input words, respectively. Instead of WER, the character error rate (CER) has been employed, while adhering to the same evaluation principle.
PESQ and MOS-LQO	$MOS - LQO = 0.999 + \frac{4.999 - 0.999}{1 + e^{-1.4945 \cdot PESQ + 4.6607}}$	PESQ serves as an objective technique for evaluating the perceived quality of speech [47]. The assessment involves assigning numerical scores within the range of -0.5 to 4.5. Additionally, a correlation can be established between MOS and perceptual evaluation of speech quality (PESQ) scores, giving rise to a novel evaluation metric termed the mean opinion score-listening quality objective (MOS-LQO), also identified as PESQ Rec.862.1. [22]
RTF	$RTF = \frac{\text{Total Processing Time}}{\text{Total Duration}}$	Real-time factor (RTF) serves as a standard metric to assess the processing time cost of an ASR system. It represents the average processing time required for one second of speech
RER	$\frac{(E_{\text{baseline}} - E_{\text{proposed}})}{E_{\text{baseline}}} \times 100\%$	The relative error rate (RER) expresses the percentage error rate achieved by the proposed DL model compared to the baseline. E_{baseline} is the error rate of the baseline model. E_{proposed} is the error rate of the proposed model or method.
D	$\frac{\sum_{i=1}^n \left(1 - \frac{\sum_{j=1}^n a_{ij} \cdot i-j }{\max(i-1 , i-2 , \dots, i-n)} \right)}{n}$	Diagonal centrality of an attention matrix (D) is defined as the mean value across the centrality of all its rows, where j represents the index of each column, n signifies the length of the input sequence, a_{ij} denotes the attention weight between the i -th and j -th elements of the input sequence, and $ i-j $ signifies the distance between the i -th and j -th elements of the input sequence [48].

of a word given the previous $n - 1$ words [40]:

$$P(w_n | w_{n-1}, w_{n-2}, \dots, w_{n-(n-1)}) = \frac{C(w_{n-(n-1)}, \dots, w_n)}{C(w_{n-(n-1)}, \dots, w_{n-1})} \quad (1)$$

In the context of ASR, the LM complements the AM by providing linguistic context. The combined probability from the AM and the LM helps in determining the most likely transcription for a given audio input during the decoding process. The frequently utilized LM in ASR systems is the backoff n-gram model.

**Figure 3:** Diagram illustrating the end-to-end framework for ASR.

2.2. Evaluation criteria in ASR

To assess the effectiveness and suitability of ASR techniques, researchers have employed diverse methods. Some of these encompass well-established DL metrics, including accuracy, F1-score, recall (sensitivity or true positive rate (TPR)), precision (also known as positive predictive value), and specificity (commonly referred to as true negative rate (TNR)) [41]. These metrics serve as crucial evaluation criteria for experimental outcomes, as evidenced in studies such as [20]. Additionally, there are ASR-specific metrics, which are detailed in Table 2.

2.3. ASR datasets

Various datasets have been employed in the literature for diverse ASR tasks. Table 3 presents a selection of datasets utilized for DTL-based ASR applications, along with their respective characteristics. It is important to note that the table primarily includes publicly accessible repositories. Furthermore, it is worth mentioning that certain datasets have undergone multiple updates and improvements over time, leading to their enhanced development.

3. Advanced ASR methods and applications

Traditional statistical LMs, such as backoff n-gram LMs, have been widely used due to their simplicity and reliability. However, bidirectional encoder representations from transformers (BERT) [60], which utilize attention models, have shown better contextual understanding compared to single-direction LMs, as demonstrated in the work of Devlin et al. [61].

In terms of AM, deep learning-based models like the deep neural network-hidden Markov model (DNN-HMM) and the connectionist temporal classification (CTC) have made significant advancements [60]. DNN-HMM models have been extensively studied in ASR research, while CTC is an end-to-end training method that does not require pre-alignment and only needs input and output sequences. The S2S model has also been successful in solving ASR tasks without using an LM or pronunciation dictionary, as described in Chiu et al. [62].

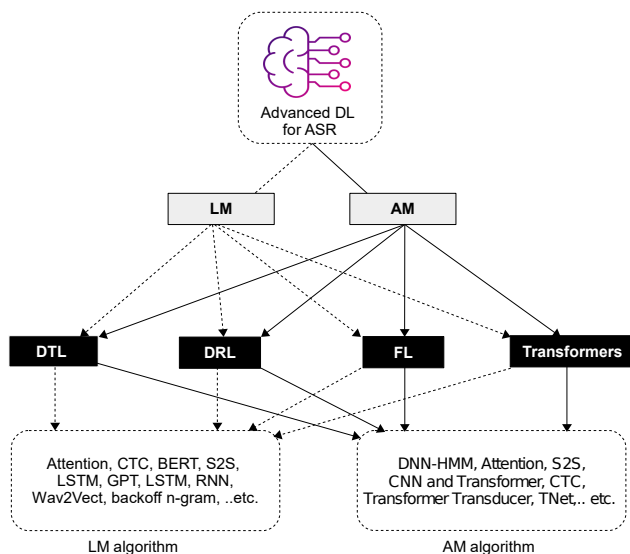
ASR systems often face performance degradation in certain situations due to the "one-model-fits-all" approach. Additionally, the lack of diverse and sufficient training data affects AM performance. To overcome these constraints and improve the resilience and flexibility of ASR systems, advanced DL methodologies such as DTL and its sub-field domain adaptation (DA), DRL, and FL have surfaced. These innovative methodologies collectively address issues concerning knowledge transfer, model generalization, and training effectiveness, offering remedies that expand upon the

Table 3

List of publicly available datasets used for advanced DL-based ASR applications

Dataset	Used by	Default ASR task	Characteristics
LibriSpeech	[49–51]	Train and assess systems for recognizing speech.	The collection consists of 1000 hours of speech recorded at a 16 kHz sampling rate, sourced from audiobooks included in the LibriVox project.
DCASE	[52, 53]	Identifying acoustic environments and detecting sound occurrences.	Comprise 8 coarse-level and 23 fine-level urban sound categories, collected in New York City in 2020 using 50 acoustic sensors.
WSJ	[48]	Acoustic scene and sound event corpus	Comprises an extensive 81 hours of meticulously curated read speech training data.
SWBD	[48]	Conversational telephone speech corpus	Is a comprehensive collection, boasting a substantial 260 hours of conversational telephone speech training data.
AISHELL	[54–56]	Chinese Mandarin speech corpus	400 participants from diverse Chinese accent regions recorded in a quiet indoor space using high-fidelity microphones, later downsampled to 16kHz.
CHIME3	[57]	SR for distant microphone in real-world settings.	Includes around 342 hours of English speech with noisy transcripts and 50 hours of noisy environment recordings.
Google-SC	[57]	Speech commands with a restricted range of words.	The dataset contains 105,829 one-second utterances of 35 words categorized by frequency. Each utterance is stored as a one-second WAVE format file with 16-bit single-channel at 16KHz rate. It involves 2,618 speakers
Aurora-4	[57]	Compare front-ends for large vocabulary recognition performance.	Aurora-4 is a speech recognition dataset derived from the WSJ corpus, offering four conditions (Clean, channel, noisy, channel+noisy) with two microphone types and six noise types, totaling 4,620 utterances per set.
Car-env	[57]	Vehicle environment sound	Is a dataset from Korea that spans 100 hours of recordings in a vehicle. It comprises brief commands, with an average of 1.6 words per command.
HKUST	[50, 56]	Classify Mandarin speech into standard and accented types.	Comprises roughly 149 hours of telephone conversations in Mandarin.
AudioSet	[51]	Audio event recognition	Includes 1,789,621 segments of 10 seconds each (equivalent to 4,971 hours). It consists of at least 100 instances clustered into 632 audio classes, with only 485 audio event categories clearly identified.
AWIC-19	[58]	Arabic words recognition	It comprises 770 recordings featuring isolated Arabic words.
TED2	[59]	English corpus for ASR	The dataset was first made available in May 2012, for training, it comprising 118 hours, 4 minutes, and 48 seconds of training data from 666 speakers, containing approximately 1.7 million words.

capabilities of traditional DL models within the ASR sphere. Thus, many research studies have focused on enhancing existing ASR systems by applying the aforementioned algorithms. Figure 4 provides an overview of the current SOTA advanced DL-based ASR and its most useful related schemes in both AM and LM.

**Figure 4:** Overview of advanced DL-driven ASR algorithms and their commonly utilized models.

The figures in this section are designed to fulfill two key objectives: firstly, to visually explain the principles behind advanced DL techniques, namely Transformers, DTL, FL, and DRL, as depicted in Figures 5, 8, 9, and 10, respectively; and secondly, to present practical examples of implementing widely recognized techniques. These include a convolutional neural network (CNN)-based Transformer illustrated in Figure 6, a particularly effective ASR application of source separation using a Transformer, highlighted in Figure 7, and a S2S-based ASR model depicted in Figure 11. This dual presentation provides young researchers and developers with valuable, practical insights on how to implement these advanced DL techniques in their projects. By doing so, it bridges the gap between theory and practice, encouraging the application and extension of these examples in new and innovative ways.

3.1. Transformer-based ASR

The Transformer stands as a prominent deep learning model extensively employed across diverse domains, including NLP, computer vision (CV), and speech processing. Originally conceived for machine translation as a S2S model, it has evolved to find applications in various fields. The Transformer heavily relies on the self-attention mechanism, enabling it to capture extensive dependencies in input sequences. The standard Transformer model incorporates the query-key-value (QKV) attention mechanism. In this setup, given matrix representations of queries $\mathbf{Q} \in \mathbb{R}^{N \times D_k}$,

Table 4

Summary of some proposed work in Transformer-based ASR. The symbol ($\uparrow$) denotes result increase, whereas ($\downarrow$) signifies result decrease. In cases where multiple scenarios are examined, only the top-performing outcome is mentioned.

Ref.	Based on	Speech recognition task	Transformer	AM/LM	Result with metric
[55]	CNN	Solve the problem of code-switching	Multi-head attention	LM	RER= 10.2%
[56]	VGGnet	Compress ASR parameters and speeds up the inference time	Low-rank multi-head attention	AM	CER= 13.09%
[57]	DNN-HMM	Improve ASR	Attention	AM	RER= 4.7% $\downarrow$
[63]	Emformer	Large scale ASR	Attention	AM	RERR= 26%
[64]	TRUNet	Sound source separation	TNet	AM	PESQ= 0.22 $\uparrow$
[65]	MHSA	Improve speech/ASR	D ² Net	AM	PESQ= 0.96 $\uparrow$
[58]	HMM	Improve ASR	Acoustic Encoder	AM	Acc= 96%
[66]	RNN-T	Acoustic re-scoring scenario	Transformer- Transducer	AM	Acc= 8% $\uparrow$
[67]	CTC	ASR, ST, Acoustic ED	All-in-one	AM	WER=0.3% $\uparrow$
[68]	CNN	Speech recognition with low latency, reduced frame rate, and streamability.	Transformer-Transducer	AM	WER= 3.6%
[69]	CTC alignment	Retrieve the acoustic embedding at the token level for better ASR	Attention	AM	51.2x RTF $\uparrow$ WER= 2.3%
[70]	RNN-LSTM	Improve the efficiency of end-to-end ASR	Attention	LM	CER=1.98%
[59]	CTC Alignment	Enhance the performance of end-to-end ASR	Autoregressive Transformer	AM	RTF= 0.0134 WER= 2.7%

keys $\mathbf{K} \in \mathbb{R}^{M \times D_k}$, and values $\mathbf{V} \in \mathbb{R}^{M \times D_v}$, the scaled dot-product attention is defined as formulated in Equation 2:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{D_k}} \right) \mathbf{V} \quad (2)$$

Here, N and M represent the lengths of queries and keys (or values), and D_k and D_v denote the dimensions of keys (or queries) and values. The softmax operation is applied row-wise to the matrix $\mathbf{A}$. Within the Transformer architecture, three attention mechanisms exist based on the source of queries and key-value pairs:

- **Self-attention:** In the Transformer encoder, the queries ($\mathbf{Q}$), keys ($\mathbf{K}$), and values ($\mathbf{V}$) are all equal to the outputs of the previous layer, considered as $\mathbf{X}$, or to the initial embeddings in the case of the first layer.
- **Masked Self-attention:** In the Transformer decoder, self-attention is constrained, allowing queries at each position to attend only to key-value pairs up to and including that position. This is accomplished by implementing a mask function, before normalization, on the attention matrix $\hat{\mathbf{A}} = \exp \left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{D_k}} \right)$, where illegal positions are masked out by setting $\hat{A}_{ij} = -\infty$ if $i < j$. This type of self-attention is often referred to as autoregressive or causal attention.
- **Cross-attention:** In cross-attention, queries originate from the results of the preceding (decoder) layer, while keys and values stem from the outputs of the encoder.

Numerous studies in the ASR field have introduced Transformer-based approaches, encompassing both the acoustic and language domains. In the subsequent subsections, we delve into a comprehensive review and detailed analysis of several cutting-edge techniques within each of these categories. Table 4 summarises the most recent Transformer-based ASR techniques used in AM and LM domains.

3.1.1. Acoustic domain

The study [57] reveals the Transformer model's increased susceptibility to input sparsity compared to the

CNN. The authors analyze the performance decline, attributing it to the Transformer's structural characteristics. Additionally, they introduce a novel regularization method to enhance the Transformer's resilience to input sparsity. This method directly regulates attention weights, i.e. output of Figure 5 (a), through silence label information in forced-alignment, offering the advantage of not requiring extra module training and excessive computation.

The paper [50] addresses a limitation in Transformer-based end-to-end modeling for ASR tasks, where intermediate features from multiple input streams may lack diversity. The proposed solution introduces a multi-level acoustic feature extraction framework, incorporating shallow and deep streams to capture both traditional features for classification and speaker-invariant deep features for diversity. A feature correlation-based fusion (FCF) strategy, employed to combine intermediate features across both the frequency and time domains, correlates and combines these features before feeding them into the Transformer encoder-decoder module.

The proposed masked autoencoding audio spectrogram Transformer (MAE-AST) operates solely on unmasked tokens [51], utilizing a large encoder. It concatenates mask tokens with encoder output embeddings, feeding them into a shallow decoder. Fine-tuning for downstream tasks involves using only the encoder, eliminating the decoder's reconstruction layers. MAE-AST represents a significant improvement over the self-supervised audio spectrogram transformer (SSAST) model for speech and audio classification. Addressing the high masking ratio issue, the method achieves a 3 $\times$ speedup and 2 $\times$ memory usage reduction. During downstream tasks, the approach consistently outperforms SSAST. To identify varieties of sounds types, Bai et al. [52] introduce SE-Trans, a cross-task model for environmental sound recognition, encompassing acoustic scene classification, urban sound tagging, and anomalous sound detection. Utilizing attention mechanisms and Transformer encoder modules, SE-Trans learns channel-wise relationships and temporal dependencies in acoustic features. The model incorporates FMix data augmentation, involving the creation of a binary mask from a randomly sampled complex matrix with a low-pass filter. SE-Trans achieves outstanding performance in ASR tasks, proven through evaluations on DCASE challenge databases, underscoring its robustness and versatility in environmental sound recognition.

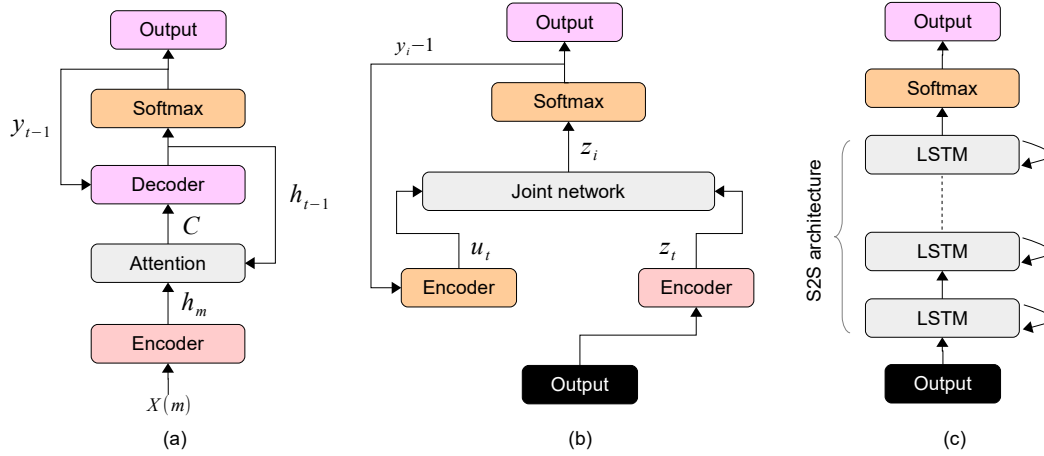


Figure 5: Three forms of end-to-end Transformers models: (a) attention, (b) RNN-Transducer, and (c) basic CTC [71].

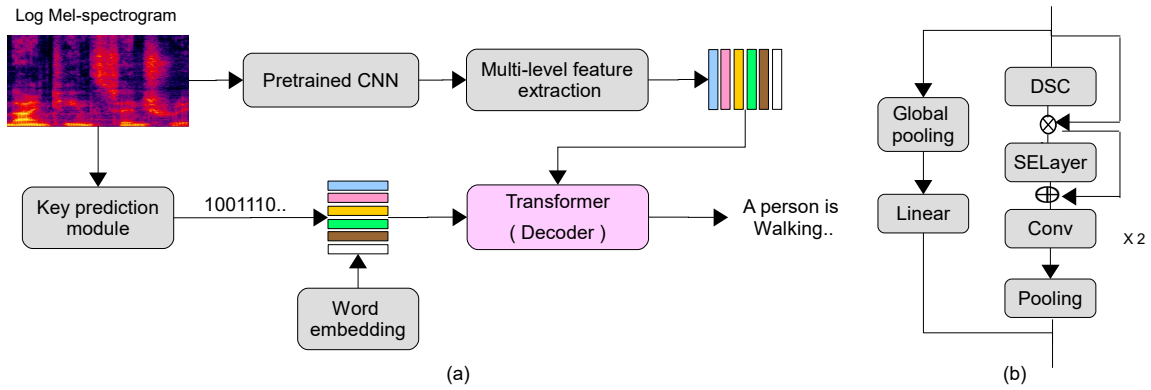


Figure 6: An example of CNN-based Transformer for automated audio captioning [53].

Automated audio captioning (AAC) involves generating textual descriptions for audio recordings, covering sound events, acoustic scenes, and event relationships. Current AAC systems typically employ an encoder-decoder architecture, with the decoder crafting captions based on extracted audio features. Chen et al. in their paper [53] introduces a novel approach that enhances caption generation by leveraging multi-level information extracted from the audio clip. The proposed method consists of a CNN encoder with multi-level feature extraction (channel attention, spatial attention), A module specialized in predicting keywords to generate guidance information at the word level and Transformer decoder. Figure 6 depicts the overall architecture incorporating the three mentioned modules. Results demonstrate significant improvements in various metrics, achieving SOTA performance during the cross-entropy training stage.

Adversarial audio involves manipulating sound to deceive or compromise machine learning (ML) systems, exploiting vulnerabilities in audio recognition models. Both [72, 73] work are built to combat adversarial noise using Transformers. The authors in [72] employed a vision Transformer customized for audio signals to identify speech regions amidst challenging acoustic conditions. To enhance adaptability, they incorporated an augmentation module as an additional head in the Transformer, integrating low-pass and band-pass filters. Experimental results reveal that the augmented vision architecture achieves an F1-score of up

to 85.2% when using a low-pass filter, surpassing the baseline vision Transformer, which attains an F1-score of up to 81.2%, in speech detection. However, in [73] the authors present an adversarial detection framework using an attention-based Transformer mechanism to identify adversarial audio. Spectrogram features are segmented and integrated with positional information before input into the Transformer encoder, achieving 96.5% accuracy under diverse conditions such as noisy environments, black-box attacks, and white-box attacks.

The paper [74] introduces a parallel-path Transformer model to address computation cost challenges for speech separation tasks. Using improved feed-forward networks and Transformer modules, it employs a parallel processing strategy with intra-chunk and inter-chunk Transformers. This enables parallel local and global modeling of speech signals, enhancing overall system performance by capturing short and long-term dependencies.

A Hybrid ASR approach outlines the conceptualization and execution of a technique that integrates neural network methodologies into advanced continuous speech recognition systems. This integration is built upon hidden Markov models (HMMs) with the aim of enhancing their overall performance. Wang et al. [75] introduce and assess Transformer-based AMs for hybrid speech recognition. The approach incorporates various positional embedding methods and an iterated loss for training deep Transformers. Demonstrating superior performance on the Librispeech benchmark, the

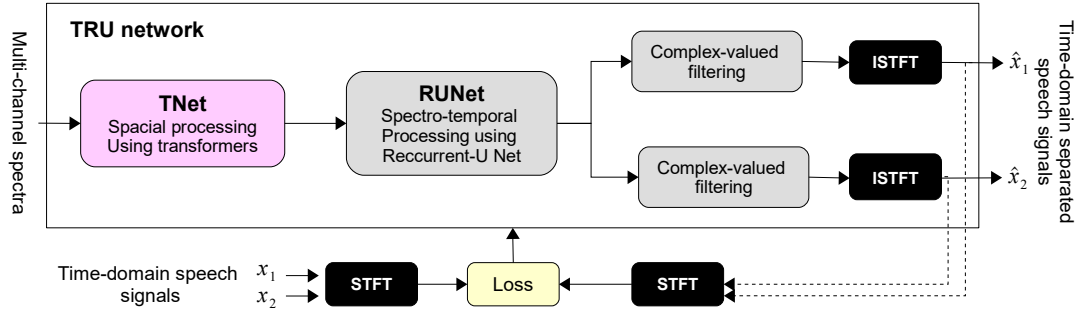


Figure 7: An example of source separation scheme based on Transformer [64].

suggested Transformer-based AM outperforms the best hybrid result by 19% to 26% relative with a standard n-gram LM.

Transformer-recurrent-U network (TRUNet) proposed in [64] represents an innovative approach to end-to-end multi-channel reverberant sound source separation. The model incorporates a recurrent-U network that directly estimates multi-channel filters from input spectra, enabling the exploitation of spatial and spectro-temporal diversity in sound source separation. In Figure 7, the block diagram illustrates the architecture of TRUNet's Transformer. The Transformer network (TNet) encompasses three variations: (i) TNet-Cat, which concatenates multi-channel spectra, treating them as a single input. This approach allows for the direct utilization of spatial information between channels. (ii) TNet-RealImag, utilizing two separate Transformer stacks for real and imaginary parts, respectively. Queries and keys are computed from the multi-channel spectra. Despite this, the method may not fully exploit spatial information, such as phase differences, directly. (iii) TNet-MagPhase, analogous to TNet-RealImag, but employing spectral magnitude and spectral phase instead of real and imaginary parts. This variation proves superior in extracting spatial information from complex-valued spectra, resulting in maximum enhanced performance in sound source separation when employing TRUNet-MagPhase architecture. Short-time Fourier transform (STFT), and inverse short-time Fourier transform (ISTFT), is to analyze and reconstruct signals in the time-frequency domain, respectively.

In recent times, dual-path networks have demonstrated effective results in many speech processing tasks such as speech separation and SE. In light of this, Wang Ke and colleagues incorporated the Transformer into the structure of dual-path networks, presenting a time-domain SE model called two-stage Transformer-based neural network (TSTNN). This model significantly enhances the performance of SE [76]. Some research findings suggest that the dot-product self-attention may not be essential for Transformer models. Similarly, the paper [65] introduces D²Net, a denoising and dereverberation network for challenging single-channel mixture speech in complex acoustic environments. D²Net incorporates a two-branch encoder (TBE) for feature extraction and fusion, along with a global-local dual-path Transformer (GLDPT) featuring local dense synthesizer attention to enhance local information perception.

The study proposed by Gong et al. [77] explores self-attention-based neural networks like the audio spectrogram

transformer (AST) for audio tasks. It introduces a self-supervised framework, improving AST performance by 60.9% on various speech classification tasks such as ASR, speaker recognition, and more, reducing reliance on labeled data. The approach marks a pioneering effort in audio self-supervised learning. Likewise, a novel augmented memory self-attention addresses limitations of Transformer-based acoustic modeling in streaming applications has been proposed in [78], outperforming existing streamable methods by over 15% in relative error reduction on benchmark datasets.

Shareef et al. in [58] propose a collaborative training method for acoustic encoders in Arabic DTL systems for speech-impaired children, achieving a 10% relative accuracy improvement on phoneme alignment in the output sequence. Pioneering in recognizing impaired children's Arabic speech. Similarly in [79], collaboratively training acoustic encoders of various sizes for on-device ASR improves efficiency and reduces redundancy. Using co-distillation with an auxiliary task, collaborative training achieves up to 11% relative WER improvement on LibriSpeech corpus.

Transducer models (Figure 5 (b)), in the context of ASR, map input sequences (acoustic features) to output sequences (transcriptions). Unlike traditional S2S models, transducers can handle variable-length input and output sequences more efficiently. The study [66] explores attention masking in Transformer-Transducer-based ASR, comparing fixed masking with chunked masking in terms of accuracy and latency. The authors claim that variable masking is the viable choice in acoustic rescoring scenarios. Similarly, to adapt the Transformer for streaming DTL, the authors in [68] employ the Transducer framework for streamable alignments. Using a unidirectional Transformer with interleaved convolution layers for audio encoding, they model future context and gradually downsample input to reduce computation cost, while limiting history context length.

Moving on, the work [67] introduces an all-in-one AM based on the Transformer architecture, combined with the CTC to ensure a sequential arrangement and utilize timing details. It addresses ASR, AT, and acoustic ED simultaneously. The model demonstrates superior performance, showcasing its suitability for comprehensive acoustic scene transcription. Winata et al. [56] propose a memory-efficient Transformer architecture for end-to-end speech recognition. It significantly reduces parameters, boosting training speed by over 50% and inference time by 1.15× compared to baseline. Experiments show better generalization, lower error

rates, and outperformance of existing schemes without external language or acoustic models. Growing demand for on-device ASR systems prompts interest in model compression.

3.1.2. Language domain

Self-attention models, such as Transformers, excel in speech recognition and reveal an important pattern. As upper self-attention layers are replaced with feed-forward layers, resembling convolutional long-short term deep neural network (CLDNN) architecture in [48], experiments on wall street journal (WSJ) and switchboard (SWBD) datasets show no performance drop and minor gains. The novel proposed metric of attention matrix diagonality indicates increased diagonality in lower to upper encoder self-attention layers. The authors conclude that a global view appears unnecessary for training upper encoder layers in speech recognition Transformers when lower layers capture sufficient contextual information. The study conducted by Hrinchuk et al. [49] presents a proficient postprocessing model for ASR with a Transformer-based encoder-decoder architecture, initialized with the weights of pre-trained BERT model [36]. The model effectively refines ASR output, demonstrating substantial performance gains through strategies like extensive data augmentation and pretrained weight initialization. On the LibriSpeech benchmark dataset, significant reductions in WERs are observed, particularly on noisier evaluation dataset portions, outperforming baseline models and approaching the performance of Transformer-XL neural language model re-scoring with 6-gram.

CTC is an architecture and principle commonly used in S2S tasks (Figure 5 (c)), such as ASR. It enables alignment-free training by introducing a blank symbol and allowing variable-length alignments between input and output sequences. During training, the model learns to align the input sequence with the target sequence, and the blank symbol accounts for multiple possible alignments. CTC is particularly effective in tasks with variable-length outputs, making it well-suited for applications like speech recognition where the duration of spoken words may vary. This latter has been used in many ASR schemes, for example, Deng et al. [54] presents the innovative pretrained Transformer S2S ASR architecture, which integrates self-supervised pretraining techniques for comprehensive end-to-end ASR. Employing a hybrid CTC/attention model, it maximizes the potential of pretrained AM and LM. The inclusion of a CTC branch aids in the encoder's convergence during training and considers all potential time boundaries in beam searching. The encoder is initiated with wav2vec2.0, and the introduction of a one-cross decoder mitigates reliance on acoustic representations, enabling initialization with pretrained DistilGPT2 and overcoming the constraint of conditioning on acoustic features.

Code-switching (CS) takes place when a speaker switches between words of two or more languages within a single sentence or across sentences. Zhou et al. [55] introduces a multi-encoder-decoder Transformer, for CS problem. It employs language-specific encoders and attention mechanisms to enhance acoustic representations, pre-trained on monolingual data to address limited CS training data. Hadwan et al. research [70] employ an attention-based encoder-decoder Transformer, to enhance end-to-end ASR for the Arabic language, focusing on Qur'an recitation. The proposed model incorporates a multi-head attention mechanism and

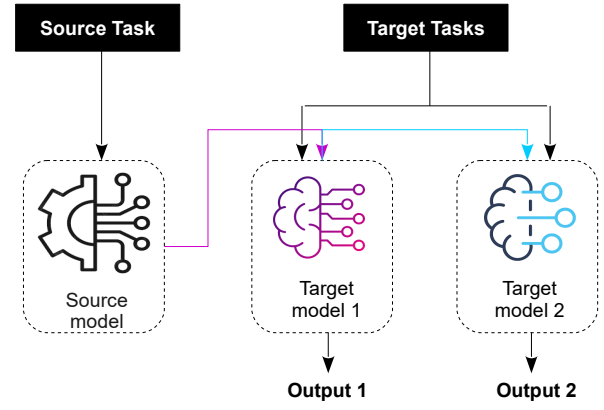


Figure 8: Deep transfer learning principle.

Mel filter bank for feature extraction. For constructing a LM, recurrent neural network (RNN) and long short term memory (LSTM) techniques were employed to train a n-gram word-based LM. The study introduces a new dataset, yielding SOTA results with a low character error rate.

3.2. DTL-based ASR

Overall, DTL consists of training a DL model on a specific domain (or task) and then transferring the acquired knowledge to a new, similar domain (or task). In what follows, we present some of the definitions that are essential to understand the principle of DTL for ASR applications.

DTL refers to a DL paradigm where knowledge gained from pre-training a model (Source model) on one domain or task is leveraged to enhance performance of target model, on a different but related domain or task. In this context, a "domain" refers to a specific data distribution, while a "task" represents a learning objective. *DA* in DTL involves adapting a model trained on a D_S to perform well on a target domain. This is crucial when there are differences in data distributions between the two domains. *Fine-tuning* is a technique where a pre-trained model is further trained on task-specific data to improve its performance on a related task. *Cross-domain learning* extends transfer learning to scenarios where the source and target domains are distinct. *Zero-shot learning* involves training a model to recognize classes not present in the training data. *Transductive* DTL focuses on adapting a model based on a specific set of target instances. *Inductive* DTL aims to generalize knowledge across domains by training a model to handle diverse tasks and domains simultaneously [36, 80, 81]. These techniques contribute to the versatility and adaptability of DL models in various applications. Figure 8 depicted the principle of DTL techniques. Table 5 summarises the most recent DTL-based ASR techniques used in AM and LM domains.

3.2.1. Acoustic domain

Schneider et al. [82] explored unsupervised pre-training for speech recognition using wav2vec model on large unlabeled audio data. The learned representations enhanced AM training with a simple CNN optimized through noise contrastive binary classification. In [83], a source filter warping data augmentation strategy is proposed to enhance the robustness of children's speech ASR. The authors constructed

an end-to-end acoustic model using the XLS-R wav2vec 2.0 model, pre-trained in a self-supervised manner on extensive cross-lingual corpora of adult speech. The work proposed in [84] introduces a multi-dialect acoustic model employing soft-parameter-sharing multi-task learning, a transductive DTL subcategory, within the Transformer architecture. Auxiliary cross-attentions aid dialect ID recognition, providing dialect information. Adaptive cross-entropy loss automatically balances multi-task learning. Experimental results demonstrate a significant reduction in error rates compared to various single- and multi-task models on multi-dialect speech recognition and dialect ID recognition tasks. Similarly, in the realm of computer vision, CNNs models like ConvNeXt have outperformed cutting-edge Transformers, partly due to the integration of depthwise separable convolutions (DSC). DSC, which approximates regular convolutions, enhances the efficiency of CNNs in terms of time and memory usage without compromising accuracy—in some cases, even enhancing it. The study [85] introduces DSC into the pre-trained audio model family for audio classification on AudioSet (target task), demonstrating its advantages in balancing accuracy and model size. Xin et al. [86] introduce an audio pyramid Transformer with an attention tree structure, with four branches, to reduce computational complexity in fine-grained audio spectrogram processing. It proposes a DA transfer learning approach for weakly supervised sound ED, a sub-field of ASR, enhancing localization performance by aligning feature distributions between frame and clip domains with a DA detection loss.

3.2.2. Language domain

The methodology is founded on the utilization of the BERT model [87], which involves pretraining language models and demonstrates improved performance across various downstream tasks. DTL approaches for language models, specifically employed in the domain of voice recognition, are referred to as LM adaptation. These approaches aim to bridge the gap between the source distribution $\mathbb{D}_S$ and the target distribution $\mathbb{D}_T$. Song et al. [88] present L2RS approach, which relies on two main components: (i) utilizing diverse textual data from SOTA NLP models, such as BERT, and (ii) automatically determining their weights to rescore the N-best lists for ASR systems.

Recent advancements in S2S models have shown promising results for training monolingual ASR systems. The CTC and encoder-decoder models are two popular architectures for end-to-end ASR. Additionally, joint training of these architectures in a multi-task hybrid approach has been explored, demonstrating improved overall performance. For instance, the architecture illustrated in Figure 5 (a) comprises S2S layers. The encoder network of S2S consists of a series of RNNs that generate embedding vectors, while the RNN decoder utilizes these vectors to produce final results. The RNN also benefits from prior predictions ($P_i, i = 0, \dots, n$), enhancing the accuracy of subsequent predictions. Moving on, a novel DTL-based approach that enhances end-to-end speech recognition has been proposed in [89]. The novelty lies in applying DTL through multilingual training and multi-task learning at two levels. The initial stage utilizes non-negative matrix factorization, instead of a bottleneck layer, and multilingual training for high-level

feature extraction. The subsequent stage employs joint CTC-attention models on these features, where the CTC was transferred to the target attention-based model. The scheme demonstrated superior performance on TIMIT but requires testing on high-resource data. Further optimization is needed for standard end-to-end training. In addition, integrating both AM and LM methodologies has the potential to enhance or construct an effective DTL-based DTL model, as demonstrated in [46, 90–94].

3.3. FL-based ASR

FL revolutionizes AI model training by enabling collaboration without the need to share sensitive training data. Traditional centralized approaches are evolving towards decentralized models, where ML algorithms are trained collaboratively on edge devices like mobile phones, laptops, or private servers [98]. The mathematical formulation of FL focuses on training a single global model across multiple devices or nodes (clients) while keeping the data localized. The objective is to minimize a global loss function that is typically the weighted sum of the local loss functions on all clients. The standard FL problem can be formulated as:

$$\min_{\theta} F(\theta) = \min_{\theta} \sum_{k=1}^K \frac{n_k}{N} F_k(\theta) \quad (3)$$

In this context, θ denotes the parameters of the global model to be learned, K represents the total number of clients, n_k signifies the number of data samples at client k , $N = \sum_{k=1}^K n_k$ stands for the total number of data samples across all clients, and $F_k(\theta)$ indicates the local loss function computed on the data of client k . In FL, the goal is to find the global model parameters θ that minimize the global loss function $F(\theta)$, which is an aggregate of the local loss functions from all participating clients. This process typically involves iterative updates to the model parameters using algorithms like federated averaging (FedAvg), where clients compute gradients or updates based on their local data and send these updates to a central server. The server then aggregates these updates to improve the global model.

- **Horizontal federated learning (HFL):** In HFL, clients train a shared global model using their respective datasets, characterized by the same feature space but different sample spaces. Each client utilizes a local AI model, and their updates are aggregated by a central server without exposing raw data. The HFL training process involves: (1) initialization, (2) local training, (3) encryption of gradients, (4) secure aggregation, and (5) global model parameter updates. The objective function minimizes a global loss across all parties' datasets [98].
- **Vertical federated learning (VFL):** VFL trains models on datasets sharing the same sample space but having different feature spaces. Through entity data alignment (EDA) and encrypted model trained (EMT), VFL allows clients to cooperatively train models without sharing raw data. The training process involves the same steps as HFL [98].

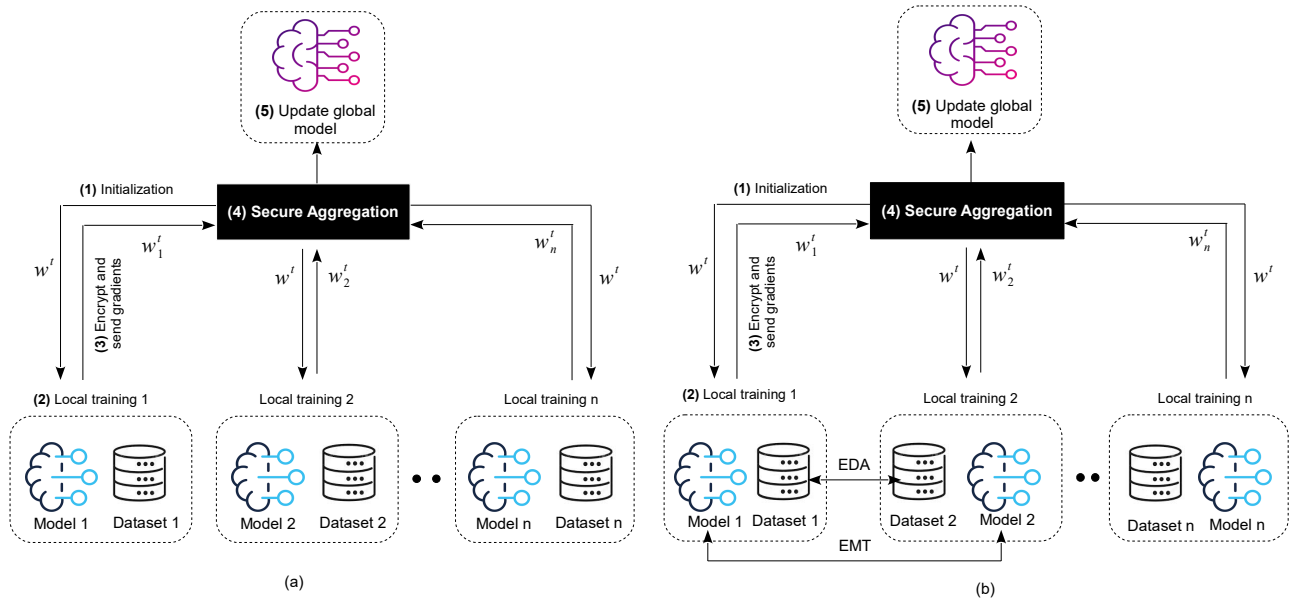
FL presents a paradigm shift in AI training, promoting collaboration while respecting data privacy. The working

Table 5

In contemporary cutting-edge frameworks, diverse pre-trained models are utilized for distinct tasks within the field. These frameworks employ different DTL approaches and assess their efficacy using specific metrics. The symbol ($\uparrow$) result increase, whereas ($\downarrow$) signifies result decrease. In cases where multiple scenarios are examined, only the top-performing outcome is mentioned.

Scheme	Based on	Speech recognition task (T_T)	AM/LM	Adaptation	Result with metric
[85]	ConvNeXt-Tiny	Audio classification	AM	DA	mAP= 0.471
[86]	APT	Sound event detection	AM	DA	F1= 79.6%
[49]	BERT (Jasper)	Speech-to-text	LM	DTL	WER= 14%
[54]	DistilGPT2	Improve ASR	Both	Fine-tuning	CER= 4.6%
[94]	XLRS Wave2vec	Improve ASR in low resource language	Both	Fine-tuning	5.6% WER $\downarrow$
[83]	XLRS Wave2vec	Improve ASR for children's speech	AM	DTL	WER = 4.86%
[91]	S2S	Speaker adaptation	Both	Features norm.	25.0% WER $\downarrow$
[84]	Transformer	Multi-dialect model aids recognizing diverse speech dialects effectively	AM	Multi-task learning	Acc= 100%
[95]	S2S	Enhancing the existing multilingual S2S model.	LM	DTL	4% CER $\downarrow$ 6% WER
[96]	PaSST	Audio tagging and event detection	AM	Fine-tuning	F1= 64.85%
[82]	Wav2vec	WSJ data speech	AM	affine transform.	36% WER $\downarrow$
[97]	ARoBERT	Spoken language understanding	LM	Fine-tuning	F1-score=92.56%

Abbreviations: Transformer (T)

**Figure 9:** Federated learning principle: (a) Horizontal FL, (b) Vertical FL.

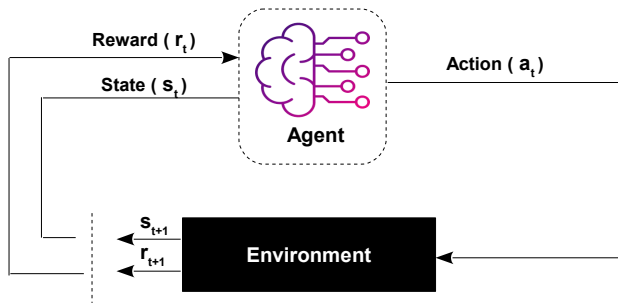
principle of HFL, and VFL is depicted on Figure 9, they cater to various data distribution scenarios, offering flexible solutions for decentralized and secure ML. The application of these FL frameworks extends across diverse domains, promising improved model accuracy and privacy preservation. The first work introducing FL in ASR is presented in [99]. The authors introduced a FL platform that is easily generalizable, incorporating hierarchical optimization and a gradient selection algorithm to enhance training time and SR performance. Guliani et al. [100] proposed a strategy to compensate non-identically distributed (non-IID) data in federated training of ASR systems. The proposed strategy involved random client data sampling, which resulted in a cost-quality trade-off. Zhu et al. [101] addressed also FL-based ASR in non-IID scenarios with personalized FL. They introduced two approaches: adapting personalization layer-based FL for ASR, involving local layers for personalized models, and proposing decoupled federated learning (DecoupleFL). DecoupleFL reduces computation on clients by shifting the computation burden to the server. Additionally, it communicates secure high-level features instead of model parameters,

reducing communication costs, particularly for large models. In [102], the authors proposed a client-adaptive federated training scheme to mitigate data heterogeneity when training ASR models. Nguyen et al. [103] used FL to train an ASR model based on a wav2vec 2.0 model pre-trained by self-supervision. Yang et al. [104] proposed a decentralized feature extraction approach in federated learning. This approach is built upon a quantum CNN (QCNN) composed of a quantum circuit encoder for feature extraction, and an RNN based end-to-end AM. This framework takes advantage of the quantum learning progress to secure models and to avoid privacy leakage attacks. Gao et al. [105] tackled a challenging and realistic ASR federated experimental setup with clients having heterogeneous data distributions, featuring thousands of different speakers, acoustic environments, and noises. Their empirical study focused on attention-based S2S end-to-end ASR models, evaluating three aggregation weighting strategies: standard FedAvg, loss-based aggregation, and a novel WER-based aggregation. Table 6 summarizes the most recent FL-based ASR techniques.

Table 6

Summary of recent proposed work in FL-based ASR. All the schemes are suggested for AM. The symbol (↑) result increase, whereas (↓) signifies result decrease. In cases where multiple scenarios are examined, only the top-performing outcome is mentioned.

Scheme	Based on	Speech recognition task	FL technique	Metric and result
[99]	S2S	Improve ASR	FedAvg	WER = 6%
[100]	End-to-end RNN-T	ASR on non-IDD data	FedAvg	WER=
[101]	CNN+Transformer extractor	ASR on non-IDD data	DecoupleFL	2.3- 3.4% WER ↓ compared with FedAvg
[102]	LSTM	ASR on non-IDD data	CAFT	WER = 15.13%
[103]	wav2vec 2.0	Improve ASR	FedAvg	WER= 10.92%
[104]	QCNN and RNN	Improve privacy-preservation in ASR	VFL	EER= 5-20%
[105]	S2S	ASR on heterogeneous data distributions	FedAvg	Accuracy = 95.12%
[106]	S2S	ASR on private and unlabelled user data.	FedNST	WER= 22.5%
[107]	Wav2vec 2.0 and Whisper	ASR & KWS for child exploitation settings	FedAvg	WER = 11-25%
[108]	Kaldi and backoff n-gram	Improve privacy-preservation in ASR	Merging models	WER= 17.7%
[109]	TDNN	Improve privacy-preservation in ASR	Aggregation	EER = 1-2%.
[110]	Non-Streaming & Streaming Conformer	Reduce client ASR model size	Federated Dropout	6-22% ↓ Client size model; 34-3% WER ↓

**Figure 10:** DRL principle.

3.4. DRL-based ASR

DRL is a ML paradigm where an agent learns optimal decision-making by interacting with an environment. The agent receives feedback in the form of rewards or penalties, adapting its behavior to maximize cumulative reward over time through a trial-and-error process. DRL involves several key concepts, as defined in the following terms: Environment model, serving as a representation of contextual dynamics; State (s), denoting the current situation perceived by the agent; Observation (o), a subset of the state directly perceived by the agent; Action (a), the decision made by the agent in response to the environment; Policy (π), describing how the agent converts environmental conditions into actions; Agent, the entity making decisions based on current states and past experiences; Reward, numerical values assigned by the environment to the agent based on state-action interactions. Figure 10 illustrates the principle of DRL.

Markov decision process (MDP) is a fundamental framework for dynamic and stochastic decision-making, characterized by state space $\mathcal{S}$, action space $\mathcal{A}$, transition probabilities $\mathbb{P}$, and a reward function R . The primary objective in an MDP is to identify an optimal policy π^* maximizing the expected discounted total reward over time, expressed as:

$$\pi^* = \max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^T \gamma^t r_t(s_t, \pi(s_t)) \right], \quad (4)$$

where γ is the discount factor. MDPs find extensive applications in addressing uncertainties in intelligent systems

within dynamic wireless environments, including spectrum management, cognitive radios, and wireless security.

In the field of ASR, DRL has primarily been proposed to tackle discrepancies between the training and testing phases. Two main discrepancies leading to potential performance deterioration have been identified: 1) The conventional use of the cross-entropy criterion maximizes log-likelihood during training, while performance is assessed by WER, not log-likelihood; 2) The teacher-forcing method, which relies on ground truth during training, implies that the model has never encountered its own predictions before testing. DRL addresses these discrepancies by bridging the gap between the training and testing phases. Several DRL-based approaches for ASR have been proposed [111–120]. For example, in [111], the authors introduced a DRL-based optimization method for the S2S ASR task called self-critical sequence training (SCST). This method can be conceptualized as a sequential decision model, depicted in Figure 11. The entire encoder-decoder neural network is treated as an agent. At each time step t , the current state s_t is formed by concatenating the acoustic feature x_t and the previous prediction Y_{t-1} . The output token serves as the action, updating the generated hypotheses sequence. SCST associates training loss and WER using a WER-related reward function, calculating the reward r_t at each token generation step by comparing it with the ground truth sequence Y^* . SCST uses the test-time beam search algorithm to sample hypotheses for reward normalization, assigning positive weights to high-reward hypotheses that outperform the current test-time system and negative weights to low-reward hypotheses. The framework is illustrated in Figure 11.

In [112], the authors developed a DRL framework for speech recognition systems using the policy gradient method. They introduced a DRL method within this framework, incorporating user feedback through hypothesis selection. Tjandra et al. [113, 114] also employed policy gradient DRL to train a S2S ASR model. In [115], the authors constructed a generic DRL-based AutoML system. This system automatically optimizes per-layer compression ratios for a SOTA attention-based end-to-end ASR model, which consists of multiple LSTM layers. The compression method employed in this work is singular value decomposition (SVD) low-rank matrix factorization. The authors

Table 7

Summary of recent proposed works in DRL-based ASR. The symbol ($\uparrow$) result increase, whereas ($\downarrow$) signifies result decrease. In cases where multiple scenarios are examined, only the top-performing outcome is mentioned.

Scheme	Model-based	ASR Tasks	DRL technique	Metric and result
[111]	S2S conformer	Improve ASR	Policy gradient	8.7%-7.8% WER $\downarrow$ over Baseline model
[112]	DNN-HMM	Improve ASR for AM	Policy gradient	WER=23.82-25.43%
[113]	S2S	Improve ASR for AM	Policy gradient	CER=6.10%
[114]	S2S	Improve ASR for AM	Policy gradient	CER=6.10%
[115]	End-to-end encoder-attention- decoder	Improve ASR Model compression for AM	Policy gradient	Up to $\sim 3\times$ compression; WER=8.06%
[116]	End-to-end encoder-attention- decoder	Improve ASR Model compression for AM	Policy gradient	Up to $\sim 5\times$ compression; WER=8.19%
[117]	CD-DNN-HMM AM & SRI LM	Speech enhancement for ASR for AM and LM	Q-learning	12.40% and 19.23% CER $\downarrow$ at 5 and 0 dB SNR conditions.
[118]	LSTM	Improve ASR for dialogue state tracking	DQN	Acc= 3.1% $\uparrow$
[119]	S2S	Improve ASR for AM	Policy gradient	CER=8.7%
[120]	Wav2vec 2.0	Improve ASR for AM	Policy gradient	4% WER $\downarrow$

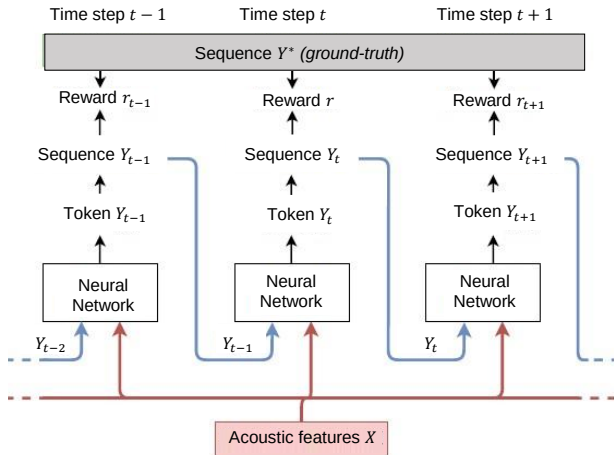


Figure 11: Example of sequential decision model of DRL-based ASR [111].

improved this approach by combining iterative compression with AutoML-based rank searching, achieving over 5 x ASR compression without degrading the WER [116]. Shen et al. [117] suggested employing DRL to optimize a SE model based on recognition results, aiming to directly enhance ASR performance. AutoML-based low-rank factorization (LRF) achieves up to 3.7 $\times$ speedup. In the shade of this, Mehrotra et al. in their work [116] propose an iterative AutoML-based LRF that employs DRL for the iterative search, surpassing 5 $\times$ compression without degrading WERs, advancing ASR. Table 7 summarizes the recent DRL-based ASR techniques.

4. Open Issues and Key challenges

Integrating advanced techniques like DTL, FL, and DRL into ASR systems presents exciting opportunities but comes with its set of challenges. This section delves into the distinct challenges associated with each approach, emphasizing the critical areas that demand attention and innovation.

4.1. Transformer-based

Transformer-based ASR systems have revolutionized the field of speech processing with their superior ability to handle sequential data. However, their deployment in the acoustic and language domains presents unique challenges and

open issues. **For the acoustic domain**, one of the key challenges in the AM is the handling of long audio sequences. Transformers require significant memory and computational resources, making it difficult to process lengthy audio inputs efficiently. Additionally, the variability in speech, such as accents, speech disorders, and background noise, can affect the robustness of transformer-based ASR systems. Developing models that can generalize across these variations without substantial data for each scenario remains an open issue. **For the language domain**, the major challenge in the LM lies in capturing the nuances of different languages and dialects. The transformer's reliance on large amounts of training data poses a challenge for low-resource languages. Furthermore, language models need to understand context deeply to accurately predict words in continuous speech, which can be particularly challenging in languages with rich morphology. **For both domains**, integrating acoustic and language models in transformer-based ASR systems to work seamlessly is complex. Achieving real-time processing speeds while maintaining high accuracy is a persistent challenge. Furthermore, the interpretability of these models is limited, making it difficult to diagnose and correct errors in ASR predictions. Addressing the trade-off between model complexity and the ability to deploy these systems on devices with limited computing power is also a critical challenge.

4.2. DTL and DA-based

This section discusses challenges and concepts related to DTL and DA in speech recognition, including distribution shift, feature space adaptation, label distribution shift, catastrophic forgetting, domain-invariant feature learning, sample selection bias, and hyperparameter optimization.

When applying a model trained on one domain (source) to another (target), a **distribution shift** often occurs, referred to as domain shift. Formally, if $P_S(X, Y)$ and $P_T(X, Y)$ represent the joint distributions of features X and labels Y in the source and target domains, respectively, the challenge arises when $P_S(X, Y) \neq P_T(X, Y)$. To address this, techniques focus on learning a transformation of the feature space to minimize the difference between the source and target distributions. This involves finding a mapping function $f : X \rightarrow Z$, where Z is a latent space in which the distributions of transformed features $f(X_S)$ and $f(X_T)$ are more similar, quantified using measures such as the maximum mean discrepancy (MMD). **Label distribution shift occurs**

when the distributions of labels ($P_S(Y)$ vs. $P_T(Y)$) differ, even if the feature distributions align. This poses challenges, especially with underrepresented classes in the target domain. Addressing this mathematically involves adjusting the model or learning process, possibly by re-weighting the loss function based on class distribution estimates. **Catastrophic forgetting** is a risk during fine-tuning on a new domain, where the model may lose its performance on the original task. Balancing loss functions (L_S for the source and L_T for the target) is crucial, often weighted by a hyperparameter λ to control their importance. **Domain-invariant feature learning** aims to learn features invariant across domains while remaining predictive. It involves optimizing a feature extractor f and predictor g to minimize the D_S loss L_S and the domain discrepancy (e.g., MMD). The problem of **sample selection bias** occurs in selecting samples for DA, affecting the effectiveness of adaptation strategies. Mathematically, addressing this bias involves weighting or selecting samples to minimize it, often using importance sampling or re-weighting techniques. **Hyperparameter optimization** is critical in DA, where the choice of hyperparameters (e.g., λ) significantly impacts performance. Finding the optimal hyperparameters typically involves solving complex optimization problems using techniques like grid search, random search, or Bayesian optimization on a validation set.

Moreover, **the unification of DTL** in ASR poses a challenge due to the varied mathematical formulations used in different studies. While efforts have been made to unify definitions and formulations, further work is needed for a consistent understanding of DTL. Speech-based DTL processing faces challenges compared to image-based processing due to potential mismatches between source and target databases arising from factors like language, speakers, age groups, ethnicity, and acoustic environments. The CTC approach, while promising, is limited by the assumption of frame independence. Cross-lingual DTL challenges include incorporating linguistic characteristics from multiple sources and integrating knowledge at different hierarchical levels, considering linguistic differences. Finally, **computational burden** remains a significant challenge in DTL and DA processes. Knowledge transfer between domains can incur additional computational costs, especially considering the extensive computational resources required for deep architectures inherent in DTL techniques.

4.3. FL-based

FL has significant potential for DTL systems, particularly in enhancing privacy and personalization. However, deploying this technology in DTL also comes with a set of challenges. Typically, in FL, data is inherently decentralized and can vary significantly across devices. This **heterogeneity in speech** data—due to differences in accents, dialects, languages, and background noise—can make it challenging to train a model that performs well across all nodes. Ensuring robustness and generalization of the DTL model under these conditions is a complex task. Moving on, FL requires periodic communication between the central server and the devices to update the model. For DTL systems, where models can be quite large, this can result in **substantial communication overhead**. Optimizing the efficiency of these updates, in terms of both bandwidth usage and energy consumption, especially on mobile devices, is

a significant challenge. Besides, although FL is designed to enhance privacy by not sharing raw data, there are still **privacy challenges**. For instance, it's possible to infer sensitive information from model updates. Ensuring that these updates do not leak private information about the users' speech data is a critical concern that requires sophisticated privacy-preserving techniques like differential privacy or secure multi-party computation.

Additionally, one of the advantages of FL is the ability to personalize models based on local data. However, **balancing personalization** with the need for a generally effective model—especially in a diverse ecosystem with varying speech patterns—is challenging. Achieving this balance without compromising the model's overall performance or the personalization benefits is a key challenge. Moving forward, FL systems need to manage potentially thousands or millions of devices participating in the training process. **Scalability issues**, including managing updates from such a large and potentially unreliable network of devices, ensuring consistent model improvements, and handling devices joining or leaving the network, are significant technical hurdles. Lastly, in FL, the distribution of data across devices is often non-IID. This means that the speech data on one device might be very different from that on another, leading to challenges in training a model that generalizes well across all devices. Overcoming the bias introduced by **non-IID data** is a major challenge in FL for ASR.

4.4. DRL-based

Using DRL in ASR systems offers promising avenues for improvement but also presents several challenges. Specifically, one of the primary challenges in applying DRL to ASR is the issue of **sparse and delayed rewards**. In many ASR tasks, the system only receives feedback (rewards or penalties) after processing lengthy sequences of speech, making it difficult to attribute the reward to specific actions or decisions. This delay complicates the learning process, as the model struggles to identify which actions led to successful outcomes. Moreover, **balancing exploration**, trying new actions to discover their effects, with exploitation, using known actions that yield the best results, is a critical challenge in DRL. In the context of ASR, this means the system must balance between adhering to known speech patterns and exploring new patterns or interpretations. Overemphasis on exploration can lead to inaccurate transcriptions, while excessive exploitation may prevent the model from adapting to new speakers or accents. Additionally, DRL models typically require a significant amount of interaction data to learn effectively. In ASR, obtaining **large volumes of labeled speech** data, especially with user feedback, can be challenging and expensive. Additionally, DRL algorithms can be sample-inefficient, meaning they need a lot of data before they start performing well, which can be a **bottleneck** in practical applications.

Moving forward, most ASR systems are built using supervised learning techniques that rely on vast amounts of annotated data. Integrating DRL into these systems poses technical challenges, as it requires a different training paradigm that focuses on learning from user interactions and feedback rather than static datasets. Besides, using DRL in ASR often involves collecting and analyzing user feedback and interactions to improve the model. This raises concerns about **user**

privacy and data security, as sensitive information might be inadvertently captured and used for training. Ensuring that data is handled securely and in compliance with privacy regulations is a significant challenge.

Designing an **appropriate reward** function that accurately reflects the desired outcomes in ASR is challenging. The reward function must capture the nuances of speech recognition, such as accuracy, naturalness, and user satisfaction, which can be difficult to quantify. Poorly designed reward functions can lead to suboptimal learning outcomes or unintended behaviors. Lastly, ASR systems are used in a wide range of environments, from quiet offices to noisy streets. DRL models need to adapt to these **varying conditions**, but training them to handle such diversity can be complex. The environment's variability requires models that can generalize well across different acoustic conditions, which remains a challenge for DRL-based ASR systems.

5. Future directions

5.1. Personalized data augmentation for dysarthric and older people

While DTL technologies have advanced significantly, especially in recognizing typical speech patterns, they still struggle to accurately identify speech from individuals with dysarthria or older adults [121]. Gathering extensive datasets from these groups is challenging due to mobility limitations often associated with these populations. In this context, personalized data augmentation plays a crucial role [122, 123]. Personalized data augmentation tailors the training process to accommodate the unique speech patterns and challenges associated with these groups. Dysarthria, a motor speech disorder, and the natural aging process can lead to speech that deviates from the normative models typically used to train DTL systems, making accurate recognition difficult [124]. Personalized data augmentation introduces a wider range of speech variations into the training dataset, including those specific to dysarthric speakers or older adults. This can include variations in speech rate, pitch, articulation, and clarity. By training on this augmented dataset, the DTL system learns to recognize and accurately transcribe speech that exhibits these characteristics [125]. Moreover, this helps the DTL models generalize better to unseen examples of speech from dysarthric speakers or older adults. This enhanced generalization is crucial for real-world applications where the system encounters a wide range of speech variations. Moving forward, personalized data augmentation can employ specific techniques tailored to the needs of dysarthric speakers or older adults, such as simulating the slurring of words, varying speech tempo, or introducing background noise [126], commonly challenging for these groups [127]. Techniques like pitch perturbation, temporal stretching, and adding noise can simulate real-world conditions more accurately for these users. A typical example is presented in [128], where a unique approach utilizes speaker-dependent generative adversarial networks (GANs) has been proposed.

5.2. Multitask learning for ASR

Multitask learning (MTL) enhances the performance of DTL systems by leveraging the inherent relatedness of multiple learning tasks to improve the generalization of the primary ASR task. This approach allows the ASR model to

learn shared representations that capture underlying patterns across different but related tasks, leading to several key benefits [129]. Typically, MTL encourages the ASR system to learn representations that are beneficial across multiple tasks. This can lead to more robust feature extraction, as the model is not optimized solely for transcribing speech but also for other related tasks, such as speaker identification or emotion recognition. This shared learning process helps in capturing a broader range of speech characteristics, which can improve the ASR system's ability to handle varied speech inputs. Moving on, by simultaneously learning related tasks, MTL acts as a form of regularization, reducing the risk of overfitting on the primary ASR task. This is because the model must find a solution that performs well across all tasks, preventing it from relying too heavily on noise or idiosyncrasies specific to the training data of the main task. Besides, learning auxiliary tasks alongside the main ASR task can improve the model's generalization capabilities. For example, learning to identify the speaker or the language can provide additional contextual clues that help the ASR system better understand and transcribe ambiguous audio signals.

Additionally, MTL can make more efficient use of available data by leveraging auxiliary tasks for which more data might be available. In scenarios where annotated data for ASR is limited, incorporating additional tasks with more abundant data can help improve the learning process and performance of the ASR system. Moreover, MTL allows ASR systems to better handle acoustic variability in speech, such as accents, dialects, or noisy environments, by incorporating tasks that directly or indirectly encourage the model to learn features that are invariant to these variations. Last but not least, modern ASR systems often employ DL architectures that can benefit from end-to-end learning strategies. MTL fits naturally into this paradigm, allowing for the joint optimization of multiple objectives within a single model architecture. This can simplify the training process and reduce the need for separately trained models or handcrafted features.

5.3. Federated multi-task learning and distillation for ASR

Federated multi-task learning (FMTL) extends the concept of FL by allowing each client to learn a personalized model that addresses its specific task, while still benefiting from collaboration with other clients. This approach recognizes the heterogeneity in clients' data distributions and tasks. Mathematically and compared with FL (Equation 3), FMTL can be formulated as:

$$\min_{\theta_1, \theta_2, \dots, \theta_K} \sum_{k=1}^K F_k(\theta_k) + \lambda R(\theta_1, \theta_2, \dots, \theta_K) \quad (5)$$

Different from FL, $R(\theta_1, \theta_2, \dots, \theta_K)$ is a regularization term that encourages some form of similarity or sharing among the model parameters of different tasks, promoting collaboration among clients. λ is a regularization coefficient that balances the trade-off between fitting the local data well and collaborating with other clients. FMTL has task-specific model parameters θ_k for each client, where only a single global model parameter θ in FL.

In this regard, FMTL offers a promising approach to improving ASR systems while also enhancing privacy and security measures. This learning paradigm extends the traditional FL model by enabling the simultaneous training of multiple tasks across distributed devices or nodes, without the need to share raw data [130]. FMTL leverages data from a wide range of devices and users, each potentially offering unique speech data, accents, dialects, and noise conditions. This diversity helps in training more robust ASR models that can perform well across various speech patterns and environments [131]. By learning from many tasks simultaneously, FMTL can personalize ASR models to individual users or specific groups without compromising the model's general performance [132]. This is particularly beneficial for users with unique speech patterns, such as those with accents or speech impairments. Moreover, FMTL encourages the development of compact models that can handle multiple tasks efficiently. For ASR systems, this means that a single model can potentially perform speech recognition, speaker identification, and even emotion detection, reducing the computational overhead on client devices [133].

On the other hand, in FMTL, raw data remains on the user's device and does not need to be shared or transferred to a central server. This inherently reduces the risk of data breaches and unauthorized access, as sensitive speech data is not centralized [134]. Additionally, FMTL can be combined with differential privacy techniques to further anonymize the model updates sent from devices to the central server. This ensures that the shared information does not reveal sensitive details about the data or the user, enhancing privacy protection [135]. Moving on, the aggregation process in FMTL can be secured using cryptographic techniques, ensuring that the aggregated model updates cannot be traced back to individual users. This secure aggregation process protects user privacy while allowing the benefits of collective learning [136]. Lastly, by aggregating model updates from a wide range of tasks and users, FMTL can improve the system's robustness to malicious attempts at data poisoning. The diversity of inputs helps in diluting the impact of any adversarial data introduced to compromise the model.

Delving deeper into techniques for FL distillation (optimizing model size) within FL frameworks is essential. This exploration involves researching methods to compress neural ASR models effectively while ensuring their performance remains intact, especially tailored for edge devices with storage and computational constraints. It is imperative to investigate the trade-offs associated with reducing model size while maintaining performance metrics like WER. Developing strategies to strike a balance between downsizing models and preserving satisfactory performance levels within FL environments is crucial.

5.4. Recent DRL techniques for ASR

Exploring incremental DRL approaches [137–139] in DRL-based ASR systems, could be very interesting. This approach involves the model continuously learning from newly acquired data and dynamically adjusting its ASR functionalities over time. By incrementally updating its knowledge base, the model can enhance its performance without necessitating full retraining, thus enabling continual enhancement of ASR systems. This capability not only fosters greater resilience and adaptability in speech recognition capabilities

but also offers potential applications in scenarios where real-time adaptation to changing conditions is crucial, such as in noisy environments or with varying speaker accents. Moreover, incremental DRL can potentially lead to more efficient use of computational resources, as the model only needs to focus on learning from new data, rather than reprocessing the entire dataset. Further research in this area could unlock new possibilities for ASR systems to evolve and improve over time, ultimately enhancing their usability and effectiveness in diverse real-world settings.

Although some DTL schemes based on DRL have been proposed, there remains a notable scarcity in the application of DRL techniques to enhance DTL methods. While policy gradient and Q-learning are commonly employed, the realm of DRL encompasses various subcategories such as double deep Q-network (DDQN), actor-critic (AC), deep deterministic policy gradient (DDPG), and more [140], which hold promise for advancing DTL with innovative approaches. Researchers are encouraged to delve into these diverse DRL-based methods to further enrich the field of DTL for both AM and LM fields.

5.5. Online DTL

Online DTL combines the principles of DTL and online learning with DNNs, enabling models to adapt in real-time to new tasks or data distributions. This approach is beneficial in dynamic environments where data arrives sequentially. DL models, specifically DNNs, learn through optimizing the weights θ to minimize a loss function L , which measures the discrepancy between predicted outputs $\hat{y}$ and true outputs y : $\theta^* = \arg \min_{\theta} L(D; \theta)$. On the other hand, DTL improves learning in a new target task through the transfer of knowledge from a related source task, adapting a pre-trained model θ_S on D_S to a D_T : $\theta_T^* = \arg \min_{\theta} L(D_T; \theta_T)$. In this regard, online learning updates the model incrementally as new data (x_t, y_t) arrives:

$$\theta_{t+1} = \theta_t - \alpha_t \nabla_{\theta} L(y_t, f(x_t; \theta_t)) \quad (6)$$

Where α_t is the learning rate, and $\nabla_{\theta} L$ is the gradient of the loss with respect to θ . Moving on, online DTL integrates these concepts to continuously adapt a deep learning model to new tasks or data streams, often involving techniques such as feature extraction, fine-tuning, model adaptation, and continual learning. The adaptation process at each time step t can be viewed as $\theta_{t+1}^* = \arg \min_{\theta} L(D_{T_t}; \theta_{T_t})$. Where D_{T_t} represents the data available at time t , including new target domain data.

The approach of online DTL offers a forward-looking solution to this issue. Particularly, discrepancies in class distributions and the representation of features between the D_S and D_T amplify the complexity of online DTL [141]. To navigate the complexities mentioned, online DTL has been dissected into two primary methodologies. The first, known as homogeneous online DTL, operates on the premise of a unified feature space across both domains. Conversely, heterogeneous online DTL acknowledges the distinct feature spaces intrinsic to each domain [142]. An exemplary solution to the challenges of heterogeneous online DTL includes leveraging unlabeled instances of co-occurrence to forge a connective bridge between the D_S and D_T , facilitating

the precursor to knowledge transfer [143]. Furthering the discourse, online DTL augmented with extreme learning machines introduces a novel framework [144]. Addressing the challenge of limited data in the D_T , the technique of DTL with lag, rooted in shallow neural network embeddings, has been applied. This method ensures the continuity of knowledge transfer, notwithstanding fluctuations in the feature set.

5.6. Transformers and LLMs-based ASR

Large language models (LLMs) and Transformers represent the forefront of AI, trained on vast datasets spanning various domains, including text, speech, images, and multi-modal inputs. Despite extensive research on ASR, existing SOTA approaches often lack integration of advanced AI techniques like DRL and FL into both AM and LM domains. For example, LLM based on DTL has demonstrated significant potential for ASR tasks, particularly for both LM and AM components. The incorporation of DTL techniques into LLM facilitates the transfer of knowledge from extensive pre-training tasks to enhance ASR effectiveness. In terms of AM, fine-tuning LLM can leverage insights gained from pre-trained models exposed to sample acoustic data. This enables the AM component to grasp acoustic features like spectrograms or Mel-frequency cepstral coefficients (MFCCs) and utilize pre-trained knowledge to improve speech recognition accuracy. Through fine-tuning, the model can adjust and specialize in specific datasets or acoustic domains, leading to enhanced ASR performance.

Similarly, the LM aspect of LLM based on DTL can enhance ASR by leveraging DTL. Pre-training LLM on vast text corpora equips it with extensive language representations, aiding in addressing diverse language challenges. Fine-tuning enables adaptation to specific language characteristics, improving transcription accuracy and contextual appropriateness. Both AM and LM fine-tuning can benefit from DA, incorporating target domain data to tailor models, reducing domain mismatch, and enhancing effectiveness and generalization. Utilizing LLM for objective ASR testing and MOS evaluation involves compiling diverse datasets, fine-tuning LLM, and integrating it into the ASR system. Evaluation metrics like CER and Pearson's correlation gauge system performance, guiding further fine-tuning iterations for improved results. This iterative process ensures the ASR system's continual enhancement and accurate MOS scale generation. Researchers are invited to explore these gaps and advance the integration of more advanced AI techniques such as DRL and FL into both AM and LM domains within Transformer-based models. Additionally, there is a need for further investigation into Transformers and DTL-based ASR schemes specifically tailored for the LM domain. Closing these gaps will contribute to the development of more robust and effective language models across various applications.

6. Conclusion

In conclusion, recent advancements in deep learning have presented both challenges and opportunities for ASR. Traditional ASR systems require extensive training datasets, often including confidential information, and consume significant computational resources. However, the demand for adaptive systems capable of performing well in dynamic environments has spurred the development of advanced deep

learning techniques such as DTL, FL, and DRL, with all their variant techniques. These advanced techniques address issues related to DA, privacy preservation, and dynamic decision-making, thereby enhancing ASR performance and reducing computational costs.

This survey has provided a comprehensive review of DTL, FL, and DRL-based DTL frameworks, offering insights into the latest developments and helping researchers and professionals understand current challenges. Additionally, the integration of Transformers, powerful DL models, has been explored for their ability to capture complex dependencies in DTL sequences. By presenting a structured taxonomy and conducting critical analyses, this paper has shed light on the strengths and weaknesses of existing frameworks, as well as highlighted ongoing challenges. Moving forward, further research is needed to overcome these challenges and unlock the full potential of advanced DL techniques in DTL. Future work should focus on refining existing approaches, addressing privacy concerns in FL, improving DRL algorithms for DTL optimization, and exploring innovative ways to leverage Transformers for more efficient and accurate speech recognition. By continuing to innovate and collaborate across disciplines, we can push the boundaries of ASR technology and realize its transformative impact on various fields, including healthcare, communication, and accessibility.

References

- [1] H. Haneche, A. Ouahabi, B. Boudraa, Compressed sensing-speech coding scheme for mobile communications, *Circuits, Systems, and Signal Processing* (2021) 1–21.
- [2] B. Essaid, H. Kheddar, N. Batel, A. Lakas, M. E. Chowdhury, Advanced artificial intelligence algorithms in cochlear implants: Review of healthcare strategies, challenges, and perspectives, *arXiv preprint arXiv:2403.15442* (2024).
- [3] Y. Luo, C. Han, N. Mesgarani, Group communication with context codec for lightweight source separation, *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 29 (2021) 1752–1761.
- [4] H. Kheddar, M. Bouzid, D. Megías, Pitch and fourier magnitude based steganography for hiding 2.4 kbps melp bitstream, *IET Signal Processing* 13 (3) (2019) 396–407.
- [5] H. Kheddar, A. C. Mazari, G. H. Ilk, Speech steganography based on double approximation of lsfs parameters in amr coding, in: *2022 7th International Conference on Image and Signal Processing and their Applications (ISPA)*, IEEE, 2022, pp. 1–8.
- [6] H. Kheddar, D. Megías, M. Bouzid, Fourier magnitude-based steganography for hiding 2.4 kbps melp secret speech, in: *2018 International Conference on Applied Smart Systems (ICASS)*, IEEE, 2018, pp. 1–5.
- [7] H. Yassine, B. Bachir, K. Aziz, A secure and high robust audio watermarking system for copyright protection, *International Journal of Computer Applications* 53 (17) (2012) 33–39.
- [8] M. Yamni, H. Karmouni, M. Sayyouri, H. Qjidaa, Efficient watermarking algorithm for digital audio/speech signal, *Digital Signal Processing* 120 (2022) 103251.
- [9] H. Chen, B. D. Rouhani, F. Koushanfar, Specmark: A spectral watermarking framework for IP protection of speech recognition systems., in: *Interspeech*, 2020, pp. 2312–2316.
- [10] M. Olivieri, R. Malvermi, M. Pezzoli, M. Zanon, S. Gonzalez, F. Antonacci, A. Sarti, Audio information retrieval and musical acoustics, *IEEE Instrumentation & Measurement Magazine* 24 (7) (2021) 10–20.
- [11] E. Wold, T. Blum, D. Keislar, J. Wheaton, Content-based classification, search, and retrieval of audio, *IEEE multimedia* 3 (3) (1996) 27–36.
- [12] W. Boes, et al., Audiovisual transfer learning for audio tagging and sound event detection, *Proceedings Interspeech 2021* (2021).

- [13] Y. Tang, J. Pino, C. Wang, X. Ma, D. Genzel, A general multi-task learning framework to leverage text data for speech to text tasks, in: ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2021, pp. 6209–6213.
- [14] F. M. Plaza-del Arco, M. D. Molina-González, L. A. Ureña-López, M. T. Martín-Valdivia, Comparing pre-trained language models for spanish hate speech detection, *Expert Systems with Applications* 166 (2021) 114120.
- [15] A. C. Mazari, H. Kheddar, Deep learning-based analysis of algerian dialect dataset targeted hate speech, offensive language and cyber-bullying, *International Journal of Computing and Digital Systems* (2023).
- [16] D. Meghraoui, B. Boudraa, T. Merazi, P. G. Vilda, A novel pre-processing technique in pathologic voice detection: Application to parkinson's disease phonation, *Biomedical Signal Processing and Control* 68 (2021) 102604.
- [17] Y.-Y. Lin, W.-Z. Zheng, W. C. Chu, J.-Y. Han, Y.-H. Hung, G.-M. Ho, C.-Y. Chang, Y.-H. Lai, A speech command control-based recognition system for dysarthric patients based on deep learning technology, *Applied Sciences* 11 (6) (2021) 2477.
- [18] Y. Kumar, S. Gupta, W. Singh, A novel deep transfer learning models for recognition of birds sounds in different environment, *Soft Computing* 26 (3) (2022) 1003–1023.
- [19] S. Padi, S. O. Sadjadi, R. D. Sriram, D. Manocha, Improved speech emotion recognition using transfer learning and spectrogram augmentation, in: *Proceedings of the 2021 International Conference on Multimodal Interaction*, 2021, pp. 645–652.
- [20] Y. Himeur, M. Elnour, F. Fadli, N. Meskin, I. Petri, Y. Rezgui, F. Bensaali, A. Amra, Next-generation energy systems for sustainable smart cities: Roles of transfer learning, *Sustainable Cities and Society* (2022) 1–35.
- [21] S. Niu, Y. Liu, J. Wang, H. Song, A decade survey of transfer learning (2010–2020), *IEEE Transactions on Artificial Intelligence* 1 (2) (2020) 151–166.
- [22] H. Kheddar, D. Megías, High capacity speech steganography for the G723.1 coder based on quantised line spectral pairs interpolation and CNN auto-encoding, *Applied Intelligence* (2022) 1–19.
- [23] T. A. de Lima, M. Da Costa-Abreu, A survey on automatic speech recognition systems for portuguese language and its variations, *Computer Speech & Language* 62 (2020) 101055.
- [24] A. Singh, V. Kadyan, M. Kumar, N. Bassan, ASRoIL: a comprehensive survey for automatic speech recognition of indian languages, *Artificial Intelligence Review* 53 (2020) 3673–3704.
- [25] R. S. Arslan, N. BARIŞCI, A detailed survey of turkish automatic speech recognition, *Turkish journal of electrical engineering and computer sciences* 28 (6) (2020) 3253–3269.
- [26] A. Dhoubi, A. Othman, O. El Ghouli, M. K. Khribi, A. Al Sinani, Arabic automatic speech recognition: a systematic literature review, *Applied Sciences* 12 (17) (2022) 8898.
- [27] J. Kaur, A. Singh, V. Kadyan, Automatic speech recognition system for tonal languages: State-of-the-art survey, *Archives of Computational Methods in Engineering* 28 (2021) 1039–1068.
- [28] A. A. Abushariah, H.-N. Ting, M. B. P. Mustafa, A. S. M. Khairuddin, M. A. Abushariah, T.-P. Tan, Bilingual automatic speech recognition: A review, taxonomy and open challenges, *IEEE Access* (2022).
- [29] J. L. K. E. Fendji, D. C. Tala, B. O. Yenke, M. Atemkeng, Automatic speech recognition using limited vocabulary: A survey, *Applied Artificial Intelligence* 36 (1) (2022) 2095039.
- [30] V. Bhardwaj, M. T. Ben Othman, V. Kukreja, Y. Belkhir, M. Bajaj, B. S. Goud, A. U. Rehman, M. Shafiq, H. Hamam, Automatic speech recognition (ASR) systems for children: A systematic literature review, *Applied Sciences* 12 (9) (2022) 4419.
- [31] R. Errattahi, A. El Hannani, H. Ouahmane, Automatic speech recognition errors detection and correction: A review, *Procedia Computer Science* 128 (2018) 32–37.
- [32] H. Aldarmaki, A. Ullah, S. Ram, N. Zaki, Unsupervised automatic speech recognition: A review, *Speech Communication* 139 (2022) 76–91.
- [33] A. S. Dhanjal, W. Singh, A comprehensive survey on automatic speech recognition using neural networks, *Multimedia Tools and Applications* (2023) 1–46.
- [34] A. B. Nassif, I. Shahin, I. Attili, M. Azzeh, K. Shaalan, Speech recognition using deep neural networks: A systematic review, *IEEE access* 7 (2019) 19143–19165.
- [35] M. Malik, M. K. Malik, K. Mehmood, I. Makhdoom, Automatic speech recognition: a survey, *Multimedia Tools and Applications* 80 (6) (2021) 9411–9457.
- [36] H. Kheddar, Y. Himeur, S. Al-Maadeed, A. Amira, F. Bensaali, Deep transfer learning for automatic speech recognition: Towards better generalization, *Knowledge-Based Systems* 277 (2023) 110851.
- [37] F. Filippidou, L. Moussiades, A benchmarking of IBM, Google and wit automatic speech recognition systems, in: *IFIP International Conference on Artificial Intelligence Applications and Innovations*, Springer, 2020, pp. 73–82.
- [38] M. Suzuki, H. Sakaji, M. Hirano, K. Izumi, Constructing and analyzing domain-specific language model for financial text mining, *Information Processing & Management* 60 (2) (2023) 103194.
- [39] C.-H. H. Yang, Y. Gu, Y.-C. Liu, S. Ghosh, I. Bulyko, A. Stolcke, Generative speech recognition error correction with large language models and task-activating prompting, in: *2023 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU)*, IEEE, 2023, pp. 1–8.
- [40] Z. Dong, Q. Ding, W. Zhai, M. Zhou, A speech recognition method based on domain-specific datasets and confidence decision networks, *Sensors* 23 (13) (2023) 6036.
- [41] H. Kheddar, Y. Himeur, A. I. Awad, Deep transfer learning for intrusion detection in industrial control networks: A comprehensive review, *Journal of Network and Computer Applications* 220 (2023) 103760.
- [42] Y. Li, X. Zhang, P. Wang, X. Zhang, Y. Liu, Insight into an unsupervised two-step sparse transfer learning algorithm for speech diagnosis of parkinson's disease, *Neural Computing and Applications* (2021) 1–18.
- [43] O. Karaman, H. Çakın, A. Alhudhaif, K. Polat, Robust automated parkinson disease detection based on voice signals with transfer learning, *Expert Systems with Applications* 178 (2021) 115013.
- [44] R. A. Ramadan, Detecting adversarial attacks on audio-visual speech recognition using deep learning method, *International Journal of Speech Technology* (2021) 1–7.
- [45] Q. Yu, Y. Ma, Y. Li, Enhancing speech recognition for parkinson's disease patient using transfer learning technique, *Journal of Shanghai Jiaotong University (Science)* (2021) 1–9.
- [46] Y. Bai, J. Yi, J. Tao, Z. Tian, Z. Wen, S. Zhang, Fast end-to-end speech recognition via non-autoregressive models and cross-modal knowledge transferring from bert, *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 29 (2021) 1897–1911.
- [47] I. Recommendation, Perceptual evaluation of speech quality (PESQ): An objective method for end-to-end speech quality assessment of narrow-band telephone networks and speech codecs, *Rec. ITU-T P. 862* (2001).
- [48] S. Zhang, E. Loweimi, P. Bell, S. Renals, On the usefulness of self-attention for automatic speech recognition with transformers, in: *2021 IEEE Spoken Language Technology Workshop (SLT)*, IEEE, 2021, pp. 89–96.
- [49] O. Hrinchuk, M. Popova, B. Ginsburg, Correction of automatic speech recognition with transformer sequence-to-sequence model, in: *ICASSP 2020-2020 IEEE international conference on acoustics, speech and signal processing (ICASSP)*, IEEE, 2020, pp. 7074–7078.
- [50] J. Li, R. Su, X. Xie, L. Wang, N. Yan, A Multi-level Acoustic Feature Extraction Framework for Transformer Based End-to-End Speech Recognition, in: *Proc. Interspeech 2022*, 2022, pp. 3173–3177. doi:10.21437/Interspeech.2022-915.
- [51] A. Baade, P. Peng, D. Harwath, Mae-ast: Masked autoencoding audio spectrogram transformer, *arXiv preprint arXiv:2203.16691* (2022).
- [52] J. Bai, J. Chen, M. Wang, M. S. Ayub, Q. Yan, A squeeze-and-excitation and transformer based cross-task model for environmental sound recognition, *IEEE Transactions on Cognitive and Developmental Systems* (2022).
- [53] K. Chen, J. Wang, F. Deng, X. Wang, icnn-transformer: An improved cnn-transformer with channel-spatial attention and keyword prediction for automated audio captioning, in: *Interspeech*, 2022, pp. 4167–4171.

- [54] K. Deng, S. Cao, Y. Zhang, L. Ma, Improving hybrid CTC/Attention end-to-end speech recognition with pretrained acoustic and language models, in: 2021 IEEE Automatic Speech Recognition and Understanding Workshop (ASRU), IEEE, 2021, pp. 76–82.
- [55] X. Zhou, E. Yilmaz, Y. Long, Y. Li, H. Li, Multi-Encoder-Decoder Transformer for Code-Switching Speech Recognition, in: Proc. Interspeech 2020, 2020, pp. 1042–1046. doi:10.21437/Interspeech.2020-2488.
- [56] G. I. Winata, S. Cahyawijaya, Z. Lin, Z. Liu, P. Fung, Lightweight and efficient end-to-end speech recognition using low-rank transformer, in: ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2020, pp. 6144–6148.
- [57] M.-H. Lee, S.-E. Lee, J.-S. Seong, J.-H. Chang, H. Kwon, C. Park, Regularizing transformer-based acoustic models by penalizing attention weights for robust speech recognition, in: Proceedings of the Annual Conference of the International Speech Communication Association, Interspeech, Vol. 2022, International Speech Communication Association, 2022, pp. 56–60.
- [58] S. R. Shareef, Y. F. Mohammed, Collaborative training of acoustic encoder for recognizing the impaired children speech, in: 2022 Fifth College of Science International Conference of Recent Trends in Information Technology (CSCTIT), IEEE, 2022, pp. 79–85.
- [59] R. Fan, W. Chu, P. Chang, A. Alwan, A ctc alignment-based non-autoregressive transformer for end-to-end automatic speech recognition, IEEE/ACM Transactions on Audio, Speech, and Language Processing 31 (2023) 1436–1448.
- [60] N. Djeflal, H. Kheddar, D. Addou, A. C. Mazari, Y. Himeur, Automatic speech recognition with bert and ctc transformers: A review, in: 2023 2nd International Conference on Electronics, Energy and Measurement (IC2EM), Vol. 1, IEEE, 2023, pp. 1–8.
- [61] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, BERT: pre-training of deep bidirectional transformers for language understanding, in: North American Chapter of the Association for Computational Linguistics, 2019.
- [62] C.-C. Chiu, T. N. Sainath, Y. Wu, R. Prabhavalkar, P. Nguyen, Z. Chen, A. Kannan, R. J. Weiss, K. Rao, E. Gonina, et al., State-of-the-art speech recognition with sequence-to-sequence models, in: 2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2018, pp. 4774–4778.
- [63] Y. Wang, Y. Shi, F. Zhang, C. Wu, J. Chan, C.-F. Yeh, A. Xiao, Transformer in action: a comparative study of transformer-based acoustic models for large scale speech recognition applications, in: ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2021, pp. 6778–6782.
- [64] A. Aroudi, S. Uhlich, M. Ferras Font, TRUNet: Transformer-Recurrent-U Network for Multi-channel Reverberant Sound Source Separation, in: Proc. Interspeech 2022, 2022, pp. 911–915. doi:10.21437/Interspeech.2022-11386.
- [65] L. Wang, W. Wei, Y. Chen, Y. Hu, D 2 net: A denoising and dereverberation network based on two-branch encoder and dual-path transformer, in: 2022 Asia-Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC), IEEE, 2022, pp. 1649–1654.
- [66] P. Swietojanski, S. Braun, D. Can, T. F. Da Silva, A. Ghoshal, T. Hori, R. Hsiao, H. Mason, E. McDermott, H. Silovsky, et al., Variable attention masking for configurable transformer transducer speech recognition, in: ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2023, pp. 1–5.
- [67] N. Moritz, G. Wichern, T. Hori, J. Le Roux, All-in-one transformer: Unifying speech recognition, audio tagging, and event detection., in: Interspeech, 2020, pp. 3112–3116.
- [68] W. Huang, W. Hu, Y. T. Yeung, X. Chen, Conv-Transformer Transducer: Low Latency, Low Frame Rate, Streamable End-to-End Speech Recognition, in: Proc. Interspeech 2020, 2020, pp. 5001–5005. doi:10.21437/Interspeech.2020-2361.
- [69] R. Fan, W. Chu, P. Chang, J. Xiao, Cass-nat: Ctc alignment-based single step non-autoregressive transformer for speech recognition, in: ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2021, pp. 5889–5893.
- [70] M. Hadwan, H. A. Alsayadi, S. AL-Haggee, An end-to-end transformer-based automatic speech recognition for qur'an reciters., Computers, Materials & Continua 74 (2) (2023).
- [71] G. Ahmed, A. A. Lawaye, T. A. Mir, P. Rana, Toward developing attention-based end-to-end automatic speech recognition, in: International Conference On Innovative Computing And Communication, Springer, 2023, pp. 147–161.
- [72] L. Smietanka, T. Maka, Augmented transformer for speech detection in adverse acoustical conditions, in: 2023 Signal Processing: Algorithms, Architectures, Arrangements, and Applications (SPA), IEEE, 2023, pp. 14–18.
- [73] Y. Li, D. Luo, Adversarial audio detection method based on transformer, in: 2022 International Conference on Machine Learning and Intelligent Systems Engineering (MLISE), IEEE, 2022, pp. 77–82.
- [74] C. Wang, M. Jia, Y. Zhang, L. Li, Parallel-path transformer network for time-domain monaural speech separation, in: 2023 International Conference on Cyber-Physical Social Intelligence (ICCSI), IEEE, 2023, pp. 509–514.
- [75] Y. Wang, A. Mohamed, D. Le, C. Liu, A. Xiao, J. Mahadeokar, H. Huang, A. Tjandra, X. Zhang, F. Zhang, et al., Transformer-based acoustic modeling for hybrid speech recognition, in: ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2020, pp. 6874–6878.
- [76] K. Wang, B. He, W.-P. Zhu, Tstnn: Two-stage transformer based neural network for speech enhancement in the time domain, in: ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), IEEE, 2021, pp. 7098–7102.
- [77] Y. Gong, C.-I. Lai, Y.-A. Chung, J. Glass, Ssast: Self-supervised audio spectrogram transformer, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 36, 2022, pp. 10699–10709.
- [78] C. Wu, Y. Wang, Y. Shi, C.-F. Yeh, F. Zhang, Streaming Transformer-Based Acoustic Models Using Self-Attention with Augmented Memory, in: Proc. Interspeech 2020, 2020, pp. 2132–2136. doi:10.21437/Interspeech.2020-2079.
- [79] V. Nagaraja, Y. Shi, G. Venkatesh, O. Kalinli, M. L. Seltzer, V. Chandra, Collaborative Training of Acoustic Encoders for Speech Recognition, in: Proc. Interspeech 2021, 2021, pp. 4573–4577. doi:10.21437/Interspeech.2021-354.
- [80] Y. Himeur, S. Al-Maadeed, H. Kheddar, N. Al-Maadeed, K. Abualsaud, A. Mohamed, T. Khattab, Video surveillance using deep transfer learning and deep domain adaptation: Towards better generalization, Engineering Applications of Artificial Intelligence 119 (2023) 105698.
- [81] H. Kheddar, M. Hemis, Y. Himeur, D. Megías, A. Amira, Deep learning for steganalysis of diverse data types: A review of methods, taxonomy, challenges and future directions, Neurocomputing (2024) 127528.
- [82] S. Schneider, A. Baevski, R. Collobert, M. Auli, wav2vec: Unsupervised pre-training for speech recognition, arXiv preprint arXiv:1904.05862 (2019).
- [83] J. Thienpondt, K. Demuyndt, Transfer Learning for Robust Low-Resource Children's Speech ASR with Transformers and Source-Filter Warping, in: Proc. Interspeech 2022, 2022, pp. 2213–2217. doi:10.21437/Interspeech.2022-10964.
- [84] Z. Dan, Y. Zhao, X. Bi, L. Wu, Q. Ji, Multi-task transformer with adaptive cross-entropy loss for multi-dialect speech recognition, Entropy 24 (10) (2022) 1429.
- [85] T. Pellegrini, I. Khalfaoui-Hassani, E. Labbé, T. Masquelier, Adapting a ConvNeXt Model to Audio Classification on AudioSet, in: Proc. Interspeech 2023, 2023, pp. 4169–4173. doi:10.21437/Interspeech.2023-1564.
- [86] Y. Xin, D. Yang, Y. Zou, Audio pyramid transformer with domain adaption for weakly supervised sound event detection and audio classification, in: Proc. Interspeech 2022, 2022, pp. 1546–1550.
- [87] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, BERT: pre-training of deep bidirectional transformers for language understanding, in: North American Chapter of the Association for Computational Linguistics, 2019.
- [88] Y. Song, D. Jiang, X. Zhao, Q. Xu, R. C.-W. Wong, L. Fan, Q. Yang, L2RS: a learning-to-rescore mechanism for automatic speech recognition, arXiv preprint arXiv:1910.11496 (2019).
- [89] C.-X. Qin, D. Qu, L.-H. Zhang, Towards end-to-end speech recognition with transfer learning, EURASIP Journal on Audio, Speech, and Music Processing 2018 (1) (2018) 1–9.

- [90] D. Jiang, C. Tan, J. Peng, C. Chen, X. Wu, W. Zhao, Y. Song, Y. Tong, C. Liu, Q. Xu, et al., A gdpr-compliant ecosystem for speech recognition with transfer, federated, and evolutionary learning, *ACM Transactions on Intelligent Systems and Technology (TIST)* 12 (3) (2021) 1–19.
- [91] F. Weninger, J. Andrés-Ferrer, X. Li, P. Zhan, Listen, Attend, Spell and Adapt: Speaker Adapted Sequence-to-Sequence ASR, in: *Proc. Interspeech 2019*, 2019, pp. 3805–3809. doi:10.21437/Interspeech.2019-2719.
- [92] S. Deena, M. Hasan, M. Doulaty, O. Saz, T. Hain, Recurrent neural network language model adaptation for multi-genre broadcast speech recognition and alignment, *IEEE/ACM Transactions on Audio, Speech, and Language Processing* 27 (3) (2018) 572–582.
- [93] S.-I. Ng, W. Liu, Z. Peng, S. Feng, H.-P. Huang, O. Scharenborg, T. Lee, The cuhk-tudelft system for the slt 2021 children speech recognition challenge, *arXiv preprint arXiv:2011.06239* (2020).
- [94] K. Manohar, G. G. Menon, A. Abraham, R. Rajan, A. Jayan, Automatic recognition of continuous malayalam speech using pretrained multilingual transformers, in: *2023 International Conference on Intelligent Systems for Communication, IoT and Security (ICISCOIS)*, IEEE, 2023, pp. 671–675.
- [95] J. Cho, M. K. Baskar, R. Li, M. Wiesner, S. H. Mallidi, N. Yalta, M. Karafiat, S. Watanabe, T. Hori, Multilingual sequence-to-sequence speech recognition: architecture, transfer learning, and language modeling, in: *2018 IEEE Spoken Language Technology Workshop (SLT)*, IEEE, 2018, pp. 521–527.
- [96] K. Li, Y. Song, I. McLoughlin, L. Liu, J. Li, L.-R. Dai, Fine-tuning audio spectrogram transformer with task-aware adapters for sound event detection.
- [97] C. Wang, S. Dai, Y. Wang, F. Yang, M. Qiu, K. Chen, W. Zhou, J. Huang, Arobert: An asr robust pre-trained language model for spoken language understanding, *IEEE/ACM Transactions on Audio, Speech, and Language Processing* (2022).
- [98] Y. Himeur, I. Varlamis, H. Kheddar, A. Amira, S. Atalla, Y. Singh, F. Bensaali, W. Mansoor, Federated learning for computer vision, *arXiv preprint arXiv:2308.13558* (2023).
- [99] D. Dimitriadis, R. G. Ken'ichi Kumtani, R. Gmyr, Y. Gaur, S. E. Eskimez, A federated approach in training acoustic models., in: *Interspeech*, 2020, pp. 981–985.
- [100] D. Guliani, F. Beaufays, G. Motta, Training speech recognition models with federated learning: A quality/cost framework, in: *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2021, pp. 3080–3084.
- [101] H. Zhu, J. Wang, G. Cheng, P. Zhang, Y. Yan, Decoupled Federated Learning for ASR with Non-IID Data, in: *Proc. Interspeech 2022*, 2022, pp. 2628–2632. doi:10.21437/Interspeech.2022-720.
- [102] X. Cui, S. Lu, B. Kingsbury, Federated acoustic modeling for automatic speech recognition, in: *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2021, pp. 6748–6752.
- [103] T. Nguyen, S. Mdhaffar, N. Tomashenko, J.-F. Bonastre, Y. Estève, Federated learning for asr based on wav2vec 2.0, in: *ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2023, pp. 1–5.
- [104] C.-H. H. Yang, J. Qi, S. Y.-C. Chen, P.-Y. Chen, S. M. Siniscalchi, X. Ma, C.-H. Lee, Decentralizing feature extraction with quantum convolutional neural network for automatic speech recognition, in: *ICASSP 2021-2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2021, pp. 6523–6527.
- [105] Y. Gao, T. Parcollet, S. Zaiem, J. Fernandez-Marques, P. P. de Gusmao, D. J. Beutel, N. D. Lane, End-to-end speech recognition from federated acoustic models, in: *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2022, pp. 7227–7231.
- [106] H. Mehmood, A. Dobrowolska, K. Saravanan, M. Ozay, FedNST: Federated Noisy Student Training for Automatic Speech Recognition, in: *Proc. Interspeech 2022*, 2022, pp. 1001–1005. doi:10.21437/Interspeech.2022-252.
- [107] J. C. Vásquez-Correa, A. Álvarez Muniain, Novel speech recognition systems applied to forensics within child exploitation: Wav2vec2.0 vs. whisper, *Sensors* 23 (4) (2023) 1843.
- [108] C. Tan, D. Jiang, H. Mo, J. Peng, Y. Tong, W. Zhao, C. Chen, R. Lian, Y. Song, Q. Xu, Federated acoustic model optimization for automatic speech recognition, in: *Database Systems for Advanced Applications: 25th International Conference, DASFAA 2020, Jeju, South Korea, September 24–27, 2020, Proceedings, Part III* 25, Springer, 2020, pp. 771–774.
- [109] N. Tomashenko, S. Mdhaffar, M. Tommasi, Y. Estève, J.-F. Bonastre, Privacy attacks for automatic speech recognition acoustic models in a federated learning framework, in: *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2022, pp. 6972–6976.
- [110] D. Guliani, L. Zhou, C. Ryu, T.-J. Yang, H. Zhang, Y. Xiao, F. Beaufays, G. Motta, Enabling on-device training of speech recognition models with federated dropout, in: *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2022, pp. 8757–8761.
- [111] C. Chen, Y. Hu, N. Hou, X. Qi, H. Zou, E. S. Chng, Self-critical sequence training for automatic speech recognition, in: *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2022, pp. 3688–3692.
- [112] T. Kala, T. Shinozaki, Reinforcement learning of speech recognition system based on policy gradient and hypothesis selection, in: *2018 IEEE international conference on acoustics, speech and signal processing (ICASSP)*, IEEE, 2018, pp. 5759–5763.
- [113] A. Tjandra, S. Sakti, S. Nakamura, Sequence-to-sequence asr optimization via reinforcement learning, in: *2018 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2018, pp. 5829–5833.
- [114] A. Tjandra, S. Sakti, S. Nakamura, End-to-end speech recognition sequence training with reinforcement learning, *IEEE Access* 7 (2019) 79758–79769.
- [115] Łukasz Dudziak, M. S. Abdelfattah, R. Vipperla, S. Laskaridis, N. D. Lane, ShrinkML: End-to-End ASR Model Compression Using Reinforcement Learning, in: *Proc. Interspeech 2019*, 2019, pp. 2235–2239. doi:10.21437/Interspeech.2019-2811.
- [116] A. Mehrotra, Łukasz Dudziak, J. Yeo, Y. Yoon Lee, R. Vipperla, M. S. Abdelfattah, S. Bhattacharya, S. Ishtiaq, A. G. C. Ramos, S. Lee, D. Kim, N. D. Lane, Iterative Compression of End-to-End ASR Model Using AutoML, in: *Proc. Interspeech 2020*, 2020, pp. 3361–3365. doi:10.21437/Interspeech.2020-1894.
- [117] Y.-L. Shen, C.-Y. Huang, S.-S. Wang, Y. Tsao, H.-M. Wang, T.-S. Chi, Reinforcement learning based speech enhancement for robust speech recognition, in: *ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2019, pp. 6750–6754.
- [118] R. T.-H. Tsai, C.-H. Chen, C.-K. Wu, Y.-C. Hsiao, H.-y. Lee, Using deep-q network to select candidates from n-best speech recognition hypotheses for enhancing dialogue state tracking, in: *ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2019, pp. 7375–7379.
- [119] H. Chung, H.-B. Jeon, J. G. Park, Semi-supervised training for sequence-to-sequence speech recognition using reinforcement learning, in: *2020 international joint conference on neural networks (IJCNN)*, IEEE, 2020, pp. 1–6.
- [120] Z. Chen, W. Zhang, End-to-end speech recognition with reinforcement learning, in: *Eighth International Conference on Electronic Technology and Information Science (ICETIS 2023)*, Vol. 12715, SPIE, 2023, pp. 392–398.
- [121] A. Hamza, D. Addou, H. Kheddar, Machine learning approaches for automated detection and classification of dysarthria severity, in: *2023 2nd International Conference on Electronics, Energy and Measurement (IC2EM)*, Vol. 1, IEEE, 2023, pp. 1–6.
- [122] S. Feng, B. M. Halpern, O. Kudina, O. Scharenborg, Towards inclusive automatic speech recognition, *Computer Speech & Language* 84 (2024) 101567.
- [123] Z. Zhou, J. Chen, N. Wang, L. Li, D. Wang, Adversarial data augmentation for robust speaker verification, *arXiv preprint arXiv:2402.02699* (2024).
- [124] Adastreamlite: Environment-adaptive streaming speech recognition on mobile devices, *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* 7 (4) (2024) 1–29.
- [125] J. H. Yeo, M. Kim, J. Choi, D. H. Kim, Y. M. Ro, Akvsr: Audio knowledge empowered visual speech recognition by compressing audio knowledge of a pretrained model, *IEEE Transactions on Multimedia* (2024).

- [126] N. Djeflal, D. Addou, H. Kheddar, S. A. Selouani, Noise-robust speech recognition: A comparative analysis of lstm and cnn approaches, in: 2023 2nd International Conference on Electronics, Energy and Measurement (IC2EM), Vol. 1, IEEE, 2023, pp. 1–6.
- [127] Q. Zhao, L. Yang, N. Lyu, A driver stress detection model via data augmentation based on deep convolutional recurrent neural network, *Expert Systems with Applications* 238 (2024) 122056.
- [128] Z. Jin, M. Geng, J. Deng, T. Wang, S. Hu, G. Li, X. Liu, Personalized adversarial data augmentation for dysarthric and elderly speech recognition, *IEEE/ACM Transactions on Audio, Speech, and Language Processing* (2023).
- [129] A. Brack, E. Entrup, M. Stamatakis, P. Buschermöhle, A. Hoppe, R. Ewerth, Sequential sentence classification in research papers using cross-domain multi-task learning, *International Journal on Digital Libraries* (2024) 1–24.
- [130] H. Zhang, M. Tao, Y. Shi, X. Bi, K. B. Letaief, Federated multi-task learning with non-stationary and heterogeneous data in wireless networks, *IEEE Transactions on Wireless Communications* (2023).
- [131] A. Singh, S. Chandrasekar, T. Sen, S. Saha, Federated multi-task learning for complaint identification using graph attention network, *IEEE Transactions on Artificial Intelligence* (2023).
- [132] X. Jiang, J. Zhang, L. Zhang, Fedradar: Federated multi-task transfer learning for radar-based internet of medical things, *IEEE Transactions on Network and Service Management* (2023).
- [133] B. Azadi, M. Haslgrübler, B. Anzengruber-Tanase, G. Sopidis, A. Ferscha, Robust feature representation using multi-task learning for human activity recognition, *Sensors* 24 (2) (2024) 681.
- [134] J. Ji, Z. Shu, H. Li, K. X. Lai, M. Lu, G. Jiang, W. Wang, Y. Zheng, X. Jiang, Edge-computing based knowledge distillation and multi-task learning for partial discharge recognition, *IEEE Transactions on Instrumentation and Measurement* (2024).
- [135] R. Šajina, N. Tanković, I. Ipšić, Multi-task peer-to-peer learning using an encoder-only transformer model, *Future Generation Computer Systems* 152 (2024) 170–178.
- [136] C. Ye, H. Zheng, Z. Hu, M. Zheng, Pfedsa: Personalized federated multi-task learning via similarity awareness, in: 2023 IEEE International Parallel and Distributed Processing Symposium (IPDPS), IEEE, 2023, pp. 480–488.
- [137] Z. Wang, C. Chen, D. Dong, Lifelong incremental reinforcement learning with online bayesian inference, *IEEE Transactions on Neural Networks and Learning Systems* 33 (8) (2021) 4003–4016.
- [138] J. Wang, J. Cao, S. Wang, Z. Yao, W. Li, Irda: Incremental reinforcement learning for dynamic resource allocation, *IEEE Transactions on Big Data* 8 (3) (2020) 770–783.
- [139] Z. Wang, C. Chen, H.-X. Li, D. Dong, T.-J. Tarn, Incremental reinforcement learning with prioritized sweeping for dynamic environments, *IEEE/ASME Transactions on Mechatronics* 24 (2) (2019) 621–632.
- [140] A. Gueriani, H. Kheddar, A. C. Mazari, Deep reinforcement learning for intrusion detection in iot: A survey, in: 2023 2nd International Conference on Electronics, Energy and Measurement (IC2EM), Vol. 1, IEEE, 2023, pp. 1–7.
- [141] P. Zhao, S. C. Hoi, J. Wang, B. Li, Online transfer learning, *Artificial Intelligence* 216 (2014) 76–102. doi:<https://doi.org/10.1016/j.artint.2014.06.003>.
- [142] Q. Wu, H. Wu, X. Zhou, M. Tan, Y. Xu, Y. Yan, T. Hao, Online transfer learning with multiple homogeneous or heterogeneous sources, *IEEE Transactions on Knowledge and Data Engineering* 29 (7) (2017) 1494–1507.
- [143] H. Wu, Y. Yan, Y. Ye, H. Min, M. K. Ng, Q. Wu, Online heterogeneous transfer learning by knowledge transition, *ACM Transactions on Intelligent Systems and Technology (TIST)* 10 (3) (2019) 1–19.
- [144] R. Alasbahi, X. Zheng, An online transfer learning framework with extreme learning machine for automated credit scoring, *IEEE Access* 10 (2022) 46697–46716. doi:[10.1109/ACCESS.2022.3171569](https://doi.org/10.1109/ACCESS.2022.3171569).